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INSTITUTE OF OPEN LEARNING**

**SMA 103
ANALYTICAL GEOMETRY**

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DEPARTMENT OF MATHEMATICS

CONTENTS

Chapter	Page
Chapter 1	
Coordinate System	1
Chapter 2	
The Straight Line	11
Chapter 3	
The Circle	25
Chapter 4	
Parabola	40
Chapter 5	
Ellipse	56
Chapter 6	
Hyperbola	68
Chapter 7	
Polar Coordinates	77
Solutions and Answers to Exercises	89

Introduction

Analytical geometry is the study of geometry by application of algebraic methods through the coordinate system. This involves changing a geometric problem into an algebraic problem so that it obeys certain rules of algebra. This module offers elementary techniques for solving geometrical problems analytically for undergraduates in mathematics and engineering students. In chapter 1, the coordinate system is covered. This also includes the study of distance between points and the locus of a point.

Chapter 2 covers the straight line and the various forms of its equation.

Chapter 3 deals with the circle, its relationship with lines and other circles on a plane.

Chapters 4, 5 & 6, deal with the conic sections. The concept of focus, directrix and tangent to the parabola, hyperbola and ellipse is introduced.

Chapter 7 deals with the polar coordinates. The equations of a circle, line, parabola, hyperbola and ellipse are considered in polar form.

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Chapter 1: The Coordinate System

1.0 Introduction

In this chapter, the Cartesian coordinate system is introduced. The distance between points and points which divide a line segment internally and externally are covered. The locus of a point as well as the equation of a locus is also covered. The general, point slope and normal form of the equation of the straight line are covered. Also covered is the relationship between lines.

Objectives

By the end of this chapter, students should be able to:

- (a) Find the distance between two points.
- (b) Find the point that divides a line segment internally into a given ratio.
- (c) Find the point that divides a line segment externally into a given ratio.
- (d) Plot the locus of a point given the equation of the locus.
- (e) Find the equation of locus given the geometrical condition under which the point moves.

1.1 The Cartesian plane

In this work the geometrical problems considered are those that are presented on two dimensional plane only and it is therefore important at this point discuss the plane.

Consider any plane with a point denoted by O , which we call the point of reference or the origin. Let line $X'X$ be horizontal and pass through O and line $Y'Y$ be vertical and pass through O . The horizontal line $X'X$ is referred to as the x -axis and the vertical line $Y'Y$ is the y -axis. Distance measured from the y -axis is referred to as an abscissa and it is positive if measured to the right and negative if measured to the left. A vertical distance from the x -axis is called a y -ordinate and it is positive if measured above the x -axis and negative if measured below the x -axis. Any point on the plane can be described by its abscissa and its ordinate as $P(x,y)$, for any point P . The symbol (x,y) represents the coordinates of the point, where x is the abscissa and y the ordinate and it is also referred to as the Cartesian coordinates of P .

The two axis partitions the plane into four quadrants as shown below. In the first quadrant both x and y are positive. In the second quadrant x is negative while y is positive. In the third quadrant both x and y are negative. In the fourth quadrant x is positive and y negative. The coordinates of the origin are $(0,0)$.

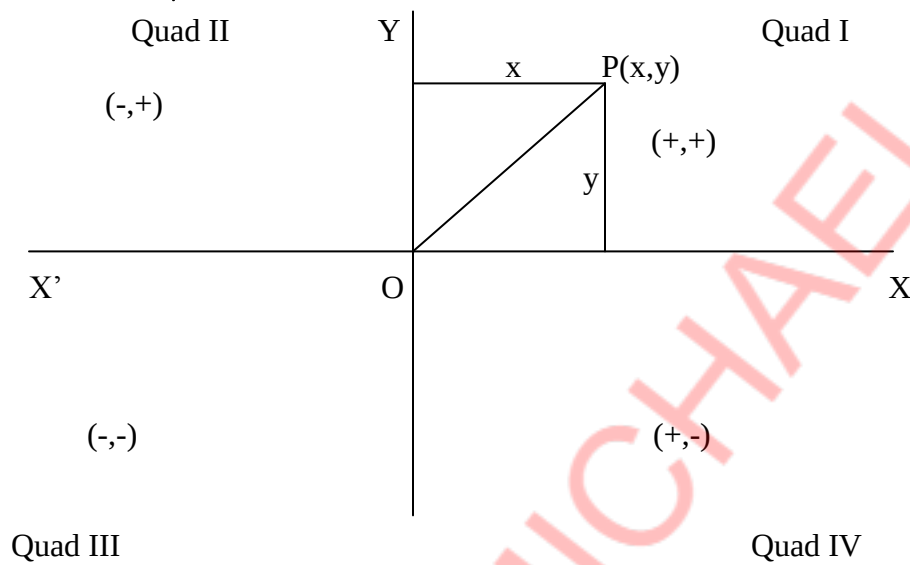


Figure1.1

1.2 Distance between two points

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be any two points on the Cartesian plane. Denote the distance between them by d_{AB} .

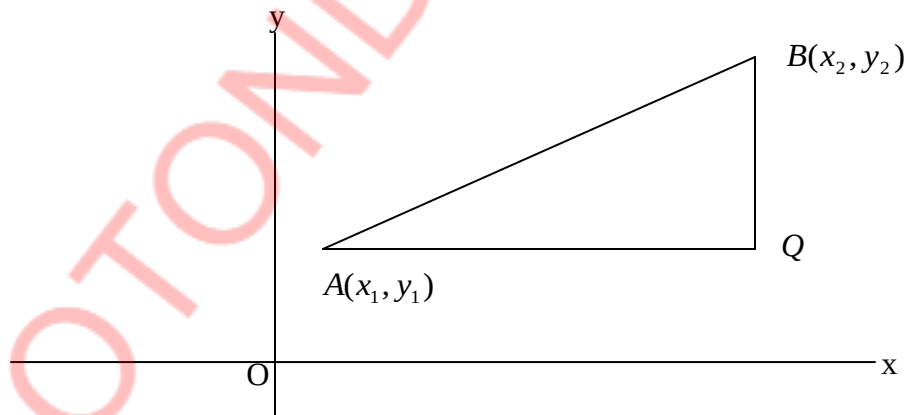


Figure1.2

Since $AQ = x_2 - x_1$, $QB = y_2 - y_1$ and $\angle AQB$ a right triangle, then

$$d_{AB} = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

Example 1.1

Find the distance between the points $A(7,9)$ and $B(-4,5)$.

Solution:

Since $x_1 = 7$, $x_2 = -4$, $y_1 = 9$ and $y_2 = 5$, the distance between the points is

$$d_{AB} = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]} = \sqrt{[(-11)^2 + (-4)^2]} = \sqrt{137} \cong 11.705$$

Example 1.2

Find the distance between the points $A(6,0)$ and $B(14,0)$.

Solution:

Since $x_1 = 6$, $x_2 = 14$, $y_1 = 0$ and $y_2 = 0$, the distance between the points is

$$d_{AB} = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]} = \sqrt{[(8)^2 + (0)^2]} = \sqrt{64} = 8$$

1.3 Points dividing a line segment

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be any two points on a line segment and $R(x, y)$ be any point between them such the ratio $AR : RB = m : n$. To get the coordinates of R, consider points Q_1 and Q_2 such that $\angle AQR$ and $\angle RQB$ are right angled triangles (figure below).

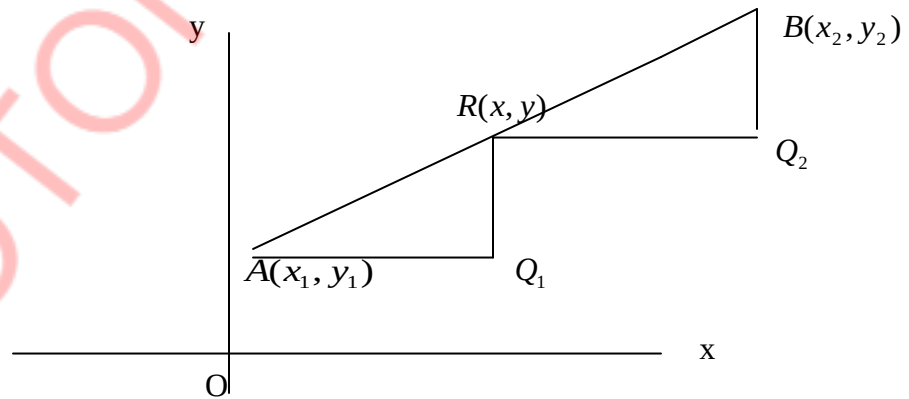


Figure1.3

Thus

$$\frac{AQ_1}{RQ_2} = \frac{Q_1R}{Q_2B} = \frac{m}{n}$$

But $AQ_1 = x - x_1$ and $RQ_2 = x_2 - x$

$\Rightarrow$

$$\frac{x - x_1}{x_2 - x} = \frac{m}{n} \Rightarrow x = \frac{nx_1 + mx_2}{m + n}$$

Similarly $Q_1R = y - y_1$ and $Q_2B = y_2 - y$ which gives

$$\frac{y - y_1}{y_2 - y} = \frac{m}{n} \Rightarrow y = \frac{ny_1 + my_2}{m + n}.$$

Hence the point which divides the line segment AB into the ratio m:n is

$$R\left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n}\right) \dots\dots\dots(1)$$

The midpoint is obtained by putting $m = n = 1$

Hence the midpoint is

$$R\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).$$

Example 1.3 Find the coordinates of the point that divides the line segment joining points A(7,-5) and b(-8,4) into the ratio 1:2.

Solution: Let the point be $R(x, y)$. Since the ratio is 1:2, then $m=1$ and $n=2$. Substituting in (1) gives

$$x = \frac{nx_1 + mx_2}{m + n} = \frac{2 \times 7 - 8}{3} = \frac{6}{3} = 2$$

and

Similarly,

$$\frac{Q_2R}{Q_3R} = \frac{y - y_1}{y - y_2} = \frac{m}{n} \Rightarrow y = \frac{my_2 - ny_1}{m - n}.$$

The point is therefore

$$R\left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n}\right)$$

Example 1.4 Find the point that divides externally the line segment joining points

- (i) A(1,1) and B(9,9) into the ratio 4:1
- (ii) A(4,-7) and B(-5,10) into the ratio 3:2.
- (iii) A(1,1) and B(9,9) into the ratio 1:4

Solution

- (i) The coordinates are $x = \frac{mx_2 - nx_1}{m - n} = \frac{4 \times 9 - 1}{3} = \frac{35}{3}$,
and

$$y = \frac{my_2 - ny_1}{m - n} = \frac{4 \times 9 - 1}{3} = \frac{35}{3}$$

and hence the point is $R(35/3, 35/3)$

- (ii) The coordinates are $x = \frac{mx_2 - nx_1}{m - n} = \frac{3 \times (-5) - 2 \times 4}{3 - 2} = \frac{-23}{1} = -23$

and

$$y = \frac{my_2 - ny_1}{m - n} = \frac{3 \times 10 - 2 \times (-7)}{3 - 2} = \frac{16}{1} = 16$$

and hence the point is $R(-23, 16)$.

- (iii) The coordinates are $x = \frac{mx_2 - nx_1}{m - n} = \frac{1 \times 7 - 4 \times 1}{1 - 4} = \frac{3}{-3} = -1$,

and

$$y = \frac{my_2 - ny_1}{m - n} = \frac{1 \times 7 - 4 \times 1}{1 - 4} = \frac{3}{-3} = -1$$

and hence the point is $R(-1, -1)$.

Examples (i) and (iii) show that the position of the division point $R(x,y)$ in relation to the points $A(x_1, y_1)$ and $B(x_2, y_2)$ is dependent on the ratio $m:n$ as in example (i). If $m > n$ then R is next to B externally and if $m < n$, R is next to A externally as in example (iii).

Exercise 1.1

- Find the distance between the following pairs of points:-
 - $(8,3)$ and $(2,7)$
 - $(-2,9)$ and $(10,12)$
 - $(6,9)$ and $(16,9)$
 - $(4,3)$ and $(-13,-20)$
- Find the lengths of the sides of the triangle whose vertices are:-
 - $(3,2)$ $(4,8)$ and $(6,1)$
 - $(-1,1)$ $(0,-9)$ and $(2,2)$
 - $(6,8)$ $(-7,3)$ and $(12,5)$
 - $(0,2)$ $(14,-8)$ and $(-5,-1)$
- Find 'b' so that the distance between $(5,2b)$ and $(8,6)$ is 5 units where b is a negative Number.
- Show that the following points lie on a straight line $(2,7)$ $(-2,15)$, $(6,-1)$
- Find 'a' so that the points $(a,-4)$ $(1,0)$ $(3,9)$ and $(-1,-2)$ form the vertices of a square.
- Find the coordinates of the points, which divide the straight line joining the given points internally in the given ratio.
 - $(-2,4)$, $(7,-3)$ ratio 4:5
 - $(7,8)$, $(12,15)$ ratio 3:2
 - $(-4,-1)$, $(8,-3)$ ratio 6:5
 - $(2,9)$, $(-4,2)$ ratio 5:7
- Find the coordinates of the points which divides the line joining the two points externally in the given ratio.
 - $(-2,1)$, $(-5,5)$ ratio 3:2
 - $(4,7)$, $(-12,13)$ ratio 8:3
 - $(9,-3)$, $(4,4)$ ratio 2:1
 - $(6,-2)$, $(-9,9)$ ratio 5:4
 - $(1,-6)$, $(10,9)$ ratio 3:7
- Find the point that divides the line segment joining the given points internally and externally.
 - $A(-6,-9)$ and $B(5,11)$ into the ratio 5:3
 - $A(-2,10)$ and $B(8,2)$ into the ratio 3:7

1.5 Locus of a point and equation of locus

Having studied the points on a plane, we now look at paths made by a moving point and how this can be described in terms of the x and y coordinates.

Definition: The locus of a point is the path traced by the point as it moves under some given geometrical condition.

Example 1.5 The locus of the point that moves such that the path it traces is at right angles with y -axis and passes through the origin is the x -axis.

Example 1.6 The locus of the point that moves such that its distance from point, say, C is a , is the circle with center C and radius a .

Definition: Equation of a locus is the mathematical function in x and y which is satisfied by all points on the locus.

As an illustration, consider the locus of the point that moves such that its distance from point, say, $C(h, k)$, is a . To get the equation of the locus, let $R(x, y)$ be any point on the locus. Then the distance between the point and the center is given by

$$\sqrt{(x-h)^2 + (y-k)^2} = a$$

This implies that the equation of circle with center $C(h, k)$ and radius a is

$$(x-h)^2 + (y-k)^2 = a^2.$$

Exercises 1.2

- 1 Find the equation of the locus of the point that moves such that its distance from point $A(3,1)$ is twice the distance from the point $B(5,1)$.
- 2 Point P moves such that its distance from points $A(-7,4)$ and $B(7,4)$ are equal. Find the equation of the locus.
- 3 Point Q moves such that its distance from the x -axis is thrice its distance from the y -axis. Find the equation of the locus.
- 4 Find the equation of the locus of the point that moves such that its distance from point $A(1,1)$ is 4 units more than its distance from point $B(5,2)$.
- 5 Plot the locus of the equation
 - (a) $2y - 3x + 1 = 0$
 - (b) $y^2 = 8x$
 - (c) $y^2 = 9 - x^2$

Miscellaneous Exercises

- Find the distance between the following pairs of points:-
 - (-2,4) and (5,8)
 - (13,-1) and (6,9)
 - (6,9) and (9,20)
 - (-18,38) and (7,-5)
- Find the lengths of the sides of the triangle whose vertices are:-
 - (-2,2) (6,6) and (9,-4)
 - (-7,2) (9,-10) and (-3,4)
 - (21,8) (-4,7) and (12,-3)
 - (21,2) (10,-8) and (6,-6)
- Find the coordinates of the points, which divide the straight line joining the given points internally in the given ratio.
 - (12,1), (0,3) ratio 3:5
 - (1,9), (6,-5) ratio 3:4
 - (-10,-4), (-8,3) ratio 1:4
 - (11,12), (-6,7) ratio 5:3
- Find the coordinates of the points which divides the line joining the two points externally in the given ratio
 - (7,5), (2,5) ratio 7:2
 - (14,-7), (8,8) ratio 9:4
 - (3,-2), (4,-8) ratio 3:7
 - (-3,7), (-2,4) ratio 2:3
 - (4,-5), (11,-5) ratio 8:7
- Find the locus of a point which moves such that its distance from the x-axis is thrice the distance from y-axis.
- Find the locus of a point that moves such that its distance from (4,3) is equivalent to distance from (7,8)
- A point moves so that its distance from the y-axis is half its distance from the origin. find the equation of locus.
- Find the point that divides the line segment joining the given points internally and externally.
 - A(5,-2) and B(6,-8) into the ratio 2:3
 - A(1,-5) and B(-8,7) into the ratio 7:3
- Find the equation of the locus of the point that moves such that its distance from point A(1,1) is 4 units more than its distance from point B(5,2) .
- Plot the locus of the equation
 - $6x - 2y + 3 = 0$
 - $x^2 = -2y$

(c) $x^2 + y^2 = 8x + 20$

Summary

In this chapter, we have introduced the concept of points on the Cartesian plane and the locus of a point. We have shown how to

- (1) Get the distance between two points,
- (2) Get the coordinates of a point that divides a line segment into a given ratio.
- (3) Plot the locus of a point given the equation of the locus.
- (4) Find the equation of locus given the geometrical condition under which the point moves

Further Reading

- 1 L. Bostock, S. Chandler, and C. Rourke. **Further Pure Mathematics**. Stanley Thornes (Publishers) Ltd, 1982.
- 2 R.E Larson and R.P. Hostetler. **Precalculus**. Houghton Mifflin Company, 1997.
- 3 P.K. Jain and Khalil Ahmad. **Analytical Geometry of Two Dimensions**. Wiley Eastern Limited, 1983.

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Chapter 2: The straight Line

2.0 Introduction

In this chapter the general, point slope and normal form of the equation of the straight line are covered. Also covered is the relationship between lines.

Objectives

By the end of this chapter, students should be able to:

- (a) Write the general, point slope and the normal form of the equation of a straight line.
- (b) Find the angle between two lines.
- (c) Find the distance between a point and a line.

2.1 The Slope and inclination of a line

In the last chapter we have referred to line segment joining two points say A and B. Through any two points a unique line passes and it extends to infinity from both ends of the line segment AB. Another way of defining a straight line is the locus of all points $R(x,y)$ such that the angle between AR and the x-axis is a constant say θ (see figure below)

In this chapter, the points and lines on the Cartesian plane are treated. The various forms of the equation of a line are considered as well as the relationships between lines.

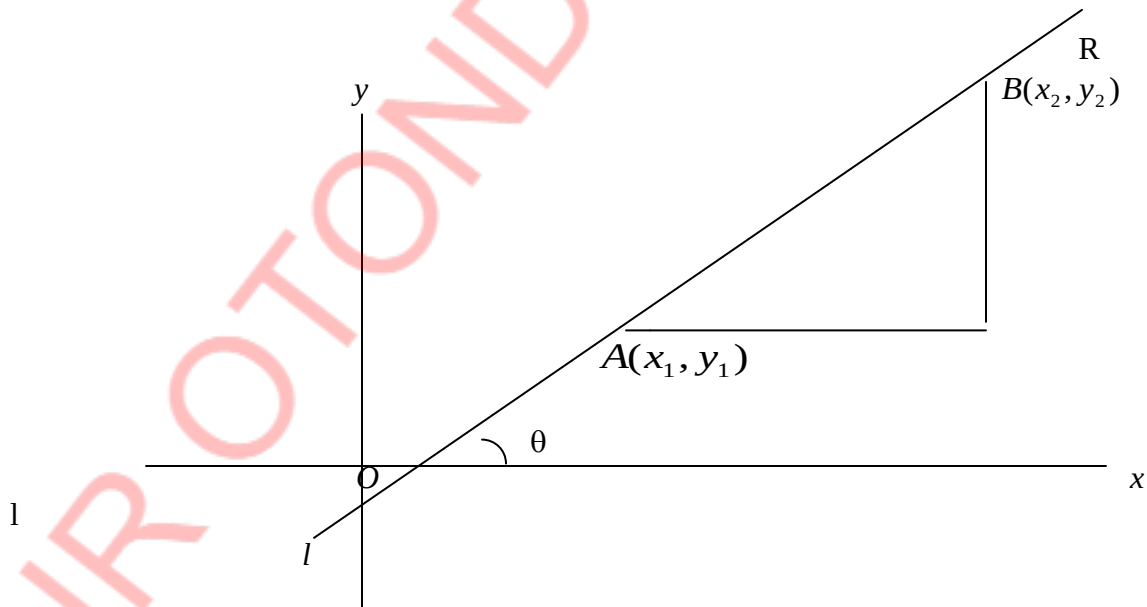


Figure2.1

The inclination of a line is the positive angle (measured anticlockwise) the line makes with the x -axis. For instance, the inclination of the line is θ .

To establish the unique relationship of all points on a line the notion of the slope is applied. Consider the two points $A(x_1, y_1)$ and $B(x_2, y_2)$ on line l in the figure above. The slope of line l is the ratio of change in y to the change in x as point $R(x, y)$ moves from A to B . Thus, the slope denoted by m is

$$m = \frac{y_2 - y_1}{x_2 - x_1} .$$

Note that the slope of a line is the trigonometric ratio of angle θ given by

$$\tan \theta = m = \frac{y_2 - y_1}{x_2 - x_1}$$

If the line is vertical, the slope is undefined.

Example 2.1

Consider the line through $(3, 8)$ and $(3, 15)$. The slope is $m = \frac{15 - 8}{3 - 3} = \frac{7}{0}$. This is undefined.

Example 2.2

Find the slope of the line passing through points $A(2, -7)$ and $B(-9, 26)$.

Solution

The slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{26 - (-7)}{-9 - 2} = \frac{33}{-11} = -3$$

Exercises 2.1

1. Find the slope of the line if defined of the straight line through each pair of points

- (a) $(-3, 8)$ and $(2, 1)$
- (b) $(14, 7)$ and $(2, 4)$
- (c) $(-7, 7)$ and $(0, 8)$
- (d) $(-12, -9)$ and $(5, 0)$
- (e) $(10, 3)$ and $(-10, -3)$
- (f) $(2, 9)$ and $(2, 11)$

2. Find 'a' so that the slope m of the line through the two points is as given.

- (a) $(4, -3)$ and $(3, 9)$ $m = 5$
- (b) $(6 - a, 2)$ and $(-1, 2)$ $m = -4$
- (c) $(2a, 4)$ and $(-7, 9a)$ $m = 3$
- a. $(3 + 2a, 5a)$ and $(1, 2a - 1)$ $m = -3$
- b. $(3a, 2)$ and $(2, 9)$ $m = 9$

2.2 Equation of straight line

Three forms of the equation of a straight line are considered here, namely, point-slope form, the general equation, and the normal equation.

2.2.1 Point-slope form.

Let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ be two points on line l such that the slope of the line is m . Let $P(x, y)$ be any other point on the line. Then

$$\frac{y - y_1}{x - x_1} = m.$$

This implies

$$y - y_1 = m(x - x_1),$$

or

$$y = mx + y_1 - mx_1$$

or

$$y = mx + c,$$

where $c = y_1 - mx_1$.

Example 2.3

Find the equation of the line through points $P_1(7, 3)$ and $P_2(-11, 9)$.

Solution

The slope of the line is

$$m = \frac{9 - 3}{-11 - 7} = \frac{-1}{3}.$$

Let $P(x, y)$ be any other point on the line. Then

$$\frac{y - 3}{x - 7} = \frac{-1}{3}$$

which reduces to

$$y = \frac{-1}{3}x + \frac{16}{3}.$$

Example 2.4

Determine whether the points $(-7, 65)$ and $(5, 23)$ lie on the line through point $(3, 5)$ whose slope is 9.

Solution

Let $P(x, y)$ be any other point on the line. Then

$$\frac{y - 5}{x - 3} = 9,$$

or

$$y = 9x - 22$$

is the equation of the line. Substituting the x coordinate of point $(-7, 65)$ in the line equation gives $y = -85$. The point therefore does not lie on the line.

Now, substituting the x coordinate of point $(5, 23)$ in the line equation gives $y = 23$. The point therefore does lie on the line.

2.2.2 Intercepts.

Intercepts on the xy plane are the directed distances from the origin to the point of intersection of the line and the coordinate axes. The y -intercept of a line is the directed distance from the origin to the point of intersection with the y -axis. At this point the x coordinate is zero. Substituting $x=0$ in the equation $y = mx + c$ gives $y = c$. This implies that when the equation of a line is in point-slope form, c gives the y -intercept.

To get the x -intercept, substitute $y=0$ in the equation and solve for x .

Example 2.5

Find the y and the x intercepts for the line through points $(-3, -28)$ and $(7, 52)$.

Solution

The slope of the line is

$$m = \frac{52 + 28}{7 + 3} = 8.$$

Let $P(x, y)$ be any other point on the line. Then

$$\frac{y + 28}{x + 3} = 8,$$

which reduces to

$$y = 8x - 4.$$

Put $x=0$ to get the y -intercept as -4 . Similarly put $y=0$ to get the x -intercept as $\frac{1}{2}$.

Exercises 2.2

1. Find the point slope form of the equation of the line which passes through the given point and given slope.

(a) $(1, 7)$ $m=4$

(b) $(-1, -2)$ $m = \frac{-2}{5}$

(c) $(9, -3)$ $m = \frac{7}{6}$

(d) $(12, 1)$ $m=-5$

$$(e) (4, -7) \quad m = \frac{-3}{7}$$

2. Find the equation in point slope form of the line passing through the given point.

- c. $(-2, -8)$ and $(4, 2)$
- d. $(3, 2)$ and $(-6, 5)$
- e. $(10, 1)$ and $(-3, -7)$
- f. $(-6, 7)$ and $(-11, -8)$
- g. $(-35, 6)$ and $(-21, -43)$

3. Find the equation of the line with intercept a on x-axis and intercept b on y-axis.

- (a) $a=2$ $b=-3$
- (b) $a=-10$ $b=-10$
- (c) $a=\frac{-2}{5}$ $b=\frac{1}{3}$
- (d) $a=\frac{2}{3}$ $b=\frac{-5}{8}$
- (e) $a=-7$ $b=\frac{-3}{7}$

2.3 General Equation of a Line

The general equation of a line is given as $Ax + By + C = 0$ where A , B and C arbitrary numbers with the exception that A and B cannot be zeros simultaneously. Note that the value C in the general equation is different from the c in the point-slope form. By making y the subject the point-slope form is obtained from the general form as

$$y = \frac{-A}{B}x + \frac{-C}{B}$$

Hence the slope is $m = \frac{-A}{B}$ and the y-intercept is $\frac{-C}{B}$.

Example 2.6

Find the general equation of the line through $(-8, 5)$ and $(21, 16)$.

Solution

First get the slope of the line which is $m = \frac{16-5}{21+8} = \frac{11}{29}$. For any point $P(x, y)$ on the line

$$\frac{y-5}{x+8} = \frac{11}{29}.$$

This reduces to

$$11x - 29y + 233 = 0$$

2.4 Normal Equation of a Line

The normal equation of a line is given in terms of the distance of the line from the origin and the angle of inclination of the line perpendicular to the line through the origin. Consider (Fig below)

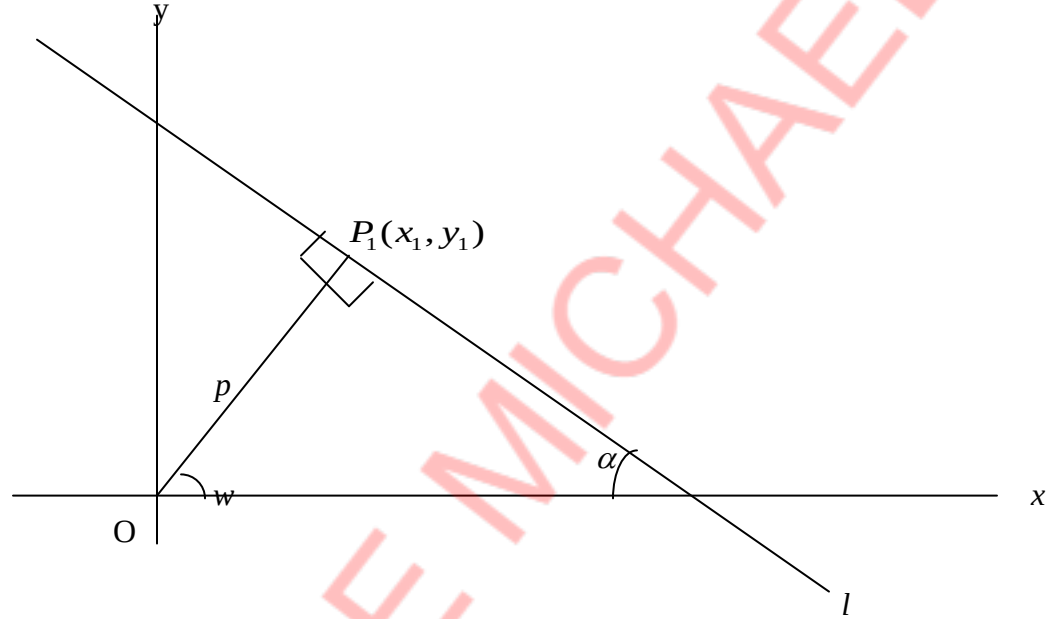


Figure 2.2: Line at distance p from origin.

Suppose the distance from the origin to the line is p . Let $P_1(x_1, y_1)$ be the point on the line such the line through O and $P_1(x_1, y_1)$ is perpendicular to l . From trigonometric ratios, we get

$$x_1 = p \cos w \text{ and } y_1 = p \sin w.$$

The slope of line through O and $P_1(x_1, y_1)$ is

$$m_2 = \frac{p \sin w}{p \cos w}.$$

The slope of l is therefore

$$m = \frac{-1}{m_2} = \frac{-p \cos w}{p \sin w} = \frac{-\cos w}{\sin w}.$$

Let $P(x, y)$ be any other point on line l . The equation of line l is

$$\frac{y - y_1}{x - x_1} = \frac{-\cos w}{\sin w},$$

or

$$y - p \sin w = \frac{-\cos w}{\sin w} (x - p \cos w),$$

which reduces to

$$x \cos w + y \sin w - p = 0.$$

This is the normal equation of a line.

Example 2.7

Find the normal equation of the line with $p=8$ and $w=30$.

Solution

Substitute the given values of p and w in (eq) to get

$$x \cos 30 + y \sin 30 - 7 = 0$$

or

$$\frac{\sqrt{3}}{2}x + \frac{1}{2}y - 7 = 0.$$

2.5 General to Normal form

Let the general form and the normal form of the equation of line l be $Ax + By + C = 0$ and $x \cos w + y \sin w - p = 0$ respectively. The corresponding coefficients in the two equations are equal or proportional. Hence

$$\frac{\cos w}{A} = \frac{\sin w}{B} = \frac{-p}{C} = k$$

$\Rightarrow$

$$\cos w = kA, \sin w = kB, -p = kC.$$

Squaring and adding the first two gives

$$\cos^2 w + \sin^2 w = k^2(A^2 + B^2) = 1.$$

$\Rightarrow$

$$k = \frac{1}{\pm \sqrt{A^2 + B^2}}.$$

The sign of the radical is taken opposite that of C , and if $C=0$, it should be the same as that of B . If $B=C=0$, the sign should be same as that of A .

Example 2.8

Find the general and normal equations to the line through point (2,5) and (3,7).

Solution

The slope of the line is $m = \frac{7-5}{3-2} = 2$. Let $P(x, y)$ be any other point on line l . Then

$\frac{y-5}{x-2} = 2$. The general equation is $2x - y + 1 = 0$. This gives $A=2$, $B=-1$ and $C=1$.

The radical is therefore $-\sqrt{A^2 + B^2} = -\sqrt{5}$. Also $\cos w = \frac{-2}{\sqrt{5}}$, $\sin w = \frac{1}{\sqrt{5}}$ and $-p = \frac{-1}{\sqrt{5}}$.

The normal equation is

$$\frac{-2}{\sqrt{5}}x + \frac{y}{\sqrt{5}} - \frac{1}{\sqrt{5}} = 0.$$

Exercise 2.3

1. Find the following equation of lines in the normal form.
 - (a) $y=7x+3$
 - (b) $5y+6x-3=0$
 - (c) $3x-9y+5=0$
 - (d) $x-4y-1=0$
 - (e) $x+8=0$
 - (f) $y-12=0$
 - (g) $6x-7y=0$
2. Find the general and normal equation of the line through the given points.
 - (a) (2,5) and (6,-1)
 - (b) (0,3) and (-9,1)
 - (c) (6,8) and (-4,7)
 - (d) (7,-8) and (10,0)
 - (e) (-11,11) and (2,-5)

2.5 Angle between two lines

Consider two lines l_1 and l_2 whose inclinations are θ_1 and θ_2 (Fig below). Their slopes are respectively $m_1 = \tan \theta_1$ and $m_2 = \tan \theta_2$. The angle between the lines is $\theta_2 - \theta_1$ and from compound angle formula,

$$\tan(\theta_2 - \theta_1) = \frac{\tan \theta_2 - \tan \theta_1}{1 + \tan \theta_1 \tan \theta_2} = \frac{m_2 - m_1}{1 + m_1 m_2}$$

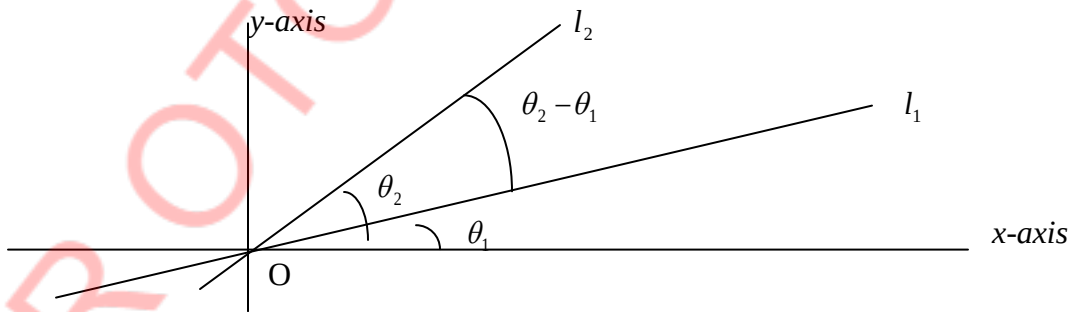


Figure 2.3: Lines with angle θ_2 between them.

Example 2.9

Find the angle between the lines $y=2x+5$ and $y=12x-1$.

Solution

The slopes of the respective lines are $m_1 = 2$ and $m_2 = 12$. Therefore

$$\tan(\theta_2 - \theta_1) = \frac{m_2 - m_1}{1 + m_1 m_2} = \frac{10}{25}.$$

The angle is therefore $\arctan(10/25) = 21.8^\circ$.

2.6 Perpendicular lines

Two lines are said to be perpendicular if the angle between them is 90° . If the slope of a given line is m_1 , then the slope of any line perpendicular to the given line is

$$m_2 = \frac{-1}{m_1}.$$

To show this, consider the two perpendicular lines l_1 and l_2 with slopes m_1 and m_2 respectively (Figure 2.4).

The slope of l_1 from the fig is $m_1 = \frac{CB}{BD}$ and that of l_2 is $m_2 = \frac{AB}{BD}$. Now, triangles $\triangle ABD$ and $\triangle DBC$ are similar and hence

$$\frac{\overline{AB}}{\overline{BD}} = \frac{\overline{BD}}{\overline{CB}}.$$

Considering directed distances, note that the change in y as point A moves to B is negative while the change is positive as C moves to B . The change in x is positive as B moves to D . Hence

$$\frac{AB}{BD} = -\frac{BD}{CB} = \frac{-1}{m_1} = m_2$$

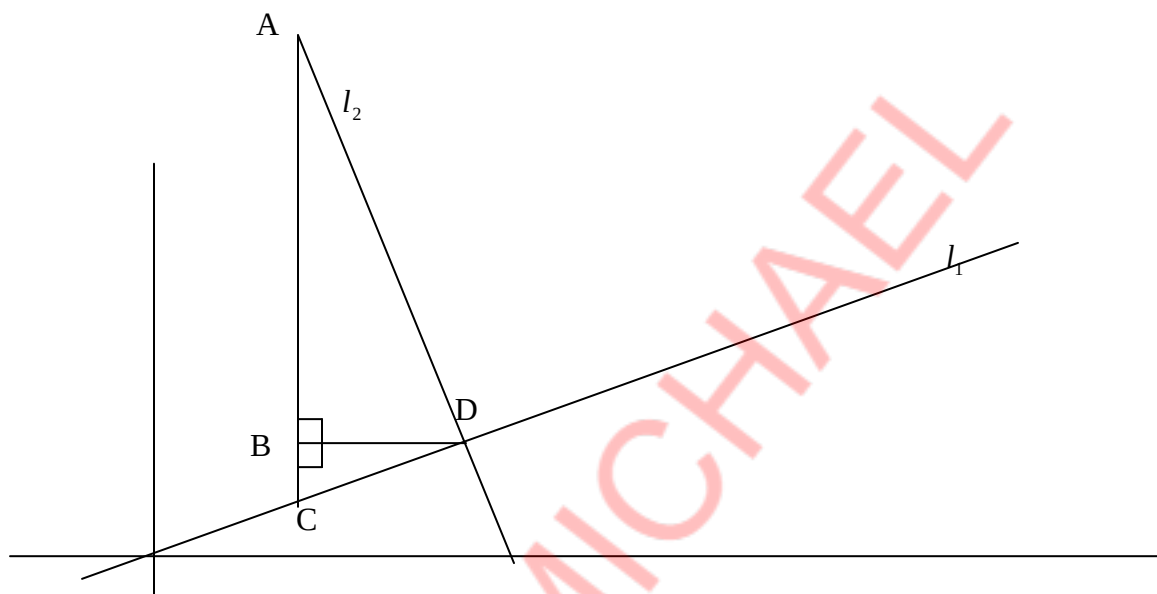


Figure 2.4: Perpendicular lines

Example 2.10

Find the equation of the line perpendicular to the line $y=5x-9$ and passes through point $(1,14)$.

Solution

The slope of the line is

$$m_2 = \frac{-1}{m_1} = \frac{-1}{5}.$$

Let $P(x, y)$ be any other point on the line. Then

$$\frac{y-14}{x-1} = \frac{-1}{5},$$

which reduces to $y = \frac{-1}{5}x + \frac{71}{5}$.

Exercises 2.4

1. Find the angle between the lines whose equations are as given.

- (a) $y=2x-3$ and $y=5x+2$
- (b) $y=-6x+5$ and $y=x-8$
- (c) $y=7x-2$ and $y=\frac{4}{3}x+11$
- (d) $y=5$ and $y=5x$
- (e) $y=-3x$ and $y=-3x+7$

(f) $y = \frac{2}{5}x$ and $y = \frac{5}{2}x + 8$

2. Find the angle between the lines, which pass through the respective, given points.

- (a) $l_1 : (2,8)$ and $(7,3)$ $l_2 : (3,6)$ and $(-6,0)$
 (b) $l_1 : (-3,2)$ and $(-1,7)$ $l_2 : (7,3)$ and $(2,5)$
 (c) $l_1 : (4,4)$ and $(6,9)$ $l_2 : (-5,7)$ and $(-3,12)$
 (d) $l_1 : (9,8)$ and $(-1,2)$ $l_2 : (2,4)$ and $(7,3)$

2.7 Distance from a point to a line

The distance from a point to a line is the length of the line segment perpendicular to the line, from the point to the line.

Suppose we wish to get the distance from point $A(m, n)$ to line l . Let l_1 be another line through A and parallel to l . Let d be the distance from A to l .

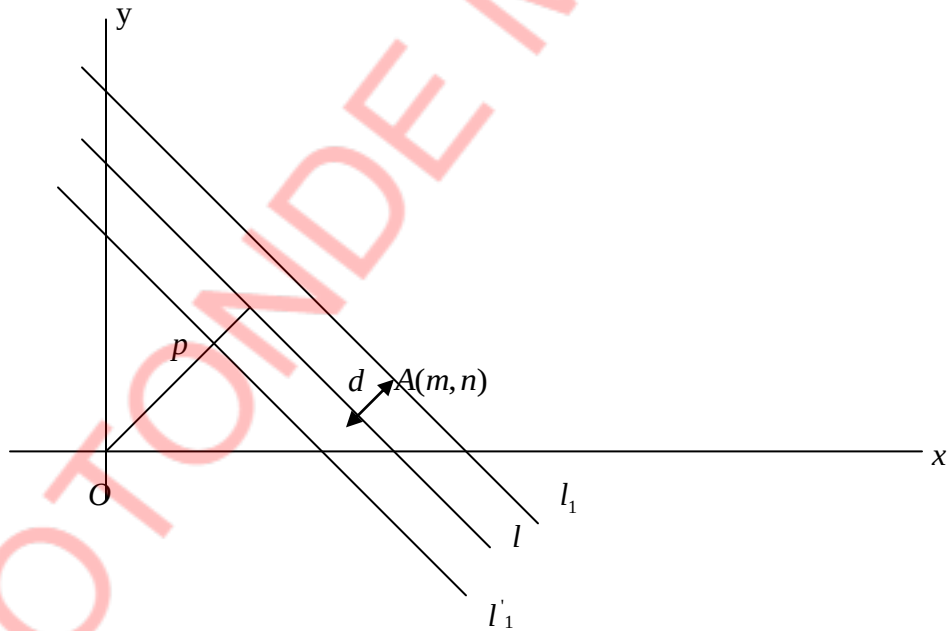


Figure 2.5: Distance between a point and a line.

Since the normal equation of l is $x \cos w + y \sin w - p = 0$, the normal equation of l_1 is $x \cos w + y \sin w - (p + d) = 0$ if l_1 is further away from the origin than l . Point $A(m, n)$ is on l_1 and hence $m \cos w + n \sin w - (p + d) = 0$. The distance is

$$d = m \cos w + n \sin w - p.$$

Suppose that line l_1 is closer to the origin than line l which implies the position of the line is now l'_1 . The equation of l'_1 is $x \cos w + y \sin w - (p - d) = 0$. Hence

$$-d = m \cos w + n \sin w - p.$$

The sign of d indicates the position of the line through $A(m, n)$ and parallel to l . If the line and the origin lie on the same side of l , the sign is negative. If they lie on the opposite sides, the sign is positive.

Example 2.11

Find the distance from the point $(4, -9)$ to the line $x \cos 60 + y \sin 60 + 5 = 0$ and state the position of the point in relation to the line and the origin.

Solution

The distance is

$$d = 4 \cos 60 - 9 \sin 60 + 5 = -0.79.$$

This implies the point and the origin lie on the same side.

Now consider the general equation $Ax + By + C = 0$. To get the distance from the point $P(x_1, y_1)$ to the line, first obtain the normal equation of the line as

$$\frac{Ax}{\sqrt{A^2 + B^2}} + \frac{By}{\sqrt{A^2 + B^2}} + \frac{C}{\sqrt{A^2 + B^2}} = 0.$$

The perpendicular distance from the point $P(x_1, y_1)$ is therefore

$$d = \frac{Ax_1 + By_1 + C}{\sqrt{A^2 + B^2}}.$$

Example 2.12

Find the distance from point $P(-2, -3)$ to the line $8x + 15y - 24 = 0$.

Solution

The normal equation of the line is

$$\frac{8x}{\sqrt{8^2 + 15^2}} + \frac{15y}{\sqrt{8^2 + 15^2}} - \frac{24}{\sqrt{8^2 + 15^2}} = 0.$$

The distance is therefore

$$\begin{aligned} d &= \frac{8(-2) + 15(-3) - 24}{\sqrt{8^2 + 15^2}} \\ &= -5. \end{aligned}$$

Exercises 2.4

2. Find the distance from the given point to the given line and state the position of the point in relation to the line and origin.
 - (a) $(2, 3)$; $y = 8 + 4x$
 - (b) $(-3, 6)$; $5x - 2y - 2 = 0$
 - (c) $(1, -7)$; $x - y + 7 = 8$
 - (d) $(10, -11)$; $-3x + 5y - 9 = 0$

- (e) $(-1,5)$; $7x+6y-30=0$
3. Let Q be a point which is equidistance from the lines $4y=2x-3$ and $y-3x-4=0$. Determine the locus of Q.

Miscellaneous Exercises

- Find the slope of the line through the given points
 - $\left(\frac{1}{2}, \frac{1}{3}\right)$ and $\left(\frac{-2}{7}, \frac{-1}{9}\right)$
 - $\left(\frac{4}{9}, \frac{-3}{10}\right)$ and $\left(\frac{1}{6}, -2\right)$
 - $\left(\frac{8}{11}, \frac{2}{7}\right)$ and $\left(4, \frac{-1}{3}\right)$
 - $\left(12, \frac{-9}{7}\right)$ and $\left(8, \frac{9}{14}\right)$
 - $(9,2)$ and $(-2,-9)$
- Find the general and normal equation of the line through the given points.
 - $(7,3)$ and $(-1,5)$
 - $(6,-8)$ and $\left(\frac{1}{2}, \frac{1}{2}\right)$
 - $(2,7)$ and $(-7,3)$
 - $(2,-1)$ and $(-1,-1)$
 - $(5,7)$ and $(5,12)$
- Find the point slope form of the equation through the given point and corresponding slope.
 - $(3,-2)$ $m=-5$
 - $\left(\frac{-11}{2}, \frac{12}{11}\right)$ $m=\frac{-2}{7}$
 - $(12,7)$ $m=\frac{7}{2}$
 - $(6,6)$ $m=\frac{-3}{5}$
 - $(18,-15)$ $m=2$
- Find the angle between the given lines.
 - $5x+8y-3=0$; $2x-4y=0$
 - $x+3y-2=0$; $-x+4y-2=0$
 - $x+8=0$; $y-9=0$
 - $2x-3y=0$; $-7x-3y=0$
 - $3x-4y=2$; $4x-3y=4$

5. Find the distance from the point to the given line.
 - (a) (4,8) ; $y = 2x + 7$
 - (b) (0,0) ; $3y = 2x + 5$
 - (c) (-7,-2) ; $x - 8 = 0$
 - (d) (3,10); $y + 2x = 0$
 - (e) (-5,-3); $4y - 3 = 0$
6. Find the point of intersection of the given lines, if they intersect.
 - (a) $3x - 7y + 22 = 0$, $y + x - 3 = 0$
 - (b) $8x - 9y - 2 = 0$, $x - 3y + 6 = 0$
 - (c) $\frac{1}{2}x + \frac{1}{3}y + 1 = 0$, $x - 3 = 0$
 - (d) $4y + 19 = 0$, $y = \frac{1}{3}x + 10 = 0$

Summary

In this chapter, have introduced the concept of points on the Cartesian plane and the straight line. We have shown how to

- (1) Determine the general, point-slope and normal form of the equation of a straight line,
- (2) Determine the angle between two lines,
- (3) Find the distance from a point to a line.

Further Reading

- 1 L. Bostock, S. Chandler, and C. Rourke. **Further Pure Mathematics.** Stanley Thornes (Publishers) Ltd, 1982.
- 2 R.E Larson and R.P. Hostetler. **Precalculus.** Houghton Mifflin Company, 1997.
- 3 P.K. Jain and Khalil Ahmad. **Analytical Geometry of Two Dimensions.** Wiley Eastern Limited, 1983.

Chapter 3: THE CIRCLE

3.0 Introduction

In this chapter we study the circle and relationships between circles. A circle is the locus of a point which moves such that its distance from a fixed point is constant. The fixed point is called the center of the circle. The distance from the point is called the radius.

Objectives

At the end of the chapter, the students should be able to:-

- (a) Determine the standard and the general equation of a circle given its:-
 - (i) Radius and the centre.
 - (ii) Any three points the circle passes through.
- (b) Find the tangent and the normal to a circle through a given point on the circle.
- (c) Find the tangents from a given point to a circle.
- (d) Determine whether circles are orthogonal.

3.1 Equation of a circle

We now consider the various forms of equations of a circle.

3.1.1 Standard Equation of circle

Consider a circle with center $C(h, k)$ and radius r (Fig)

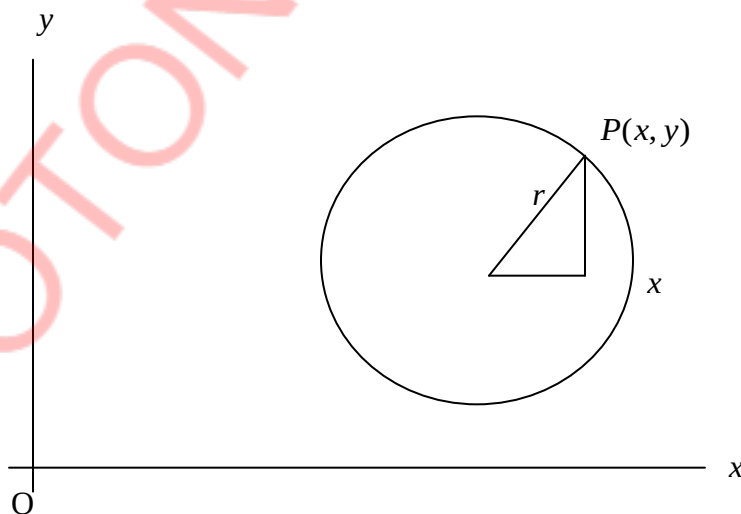


Figure 3.1

From the definition of the circle

$$\overline{CP} = r = \sqrt{(x-h)^2 + (y-k)^2}.$$

Hence the equation of the circle is

$$(x-h)^2 + (y-k)^2 = r^2.$$

If the center of the circle is at the origin the equation is

$$x^2 + y^2 = r^2.$$

Example 3.1

Find the equation of the circle with center $C(0,0)$ and radius $r=7$.

Solution

Substitute the coordinates of the center in (eq) to get

$$x^2 + y^2 = 49.$$

3.1.2 General Equation of a circle

Expanding the equation $(x-h)^2 + (y-k)^2 = r^2$ we get

$$x^2 + y^2 - 2hx - 2ky + h^2 + k^2 - r^2 = 0.$$

This can be written as

$$x^2 + y^2 + Dx + Ey + C = 0$$

where $D = -2h, E = -2k$ and $C = h^2 + k^2 - r^2$. Equation () is called the general equation of the circle.

If the equation of a circle is given in general form the center and the radius can be obtained by completing the squares as

$$\left(x + \frac{D}{2}\right)^2 + \left(y + \frac{E}{2}\right)^2 = \frac{D^2 + E^2 - 4C}{4}.$$

This is the equation of a circle with center $C\left(-\frac{D}{2}, -\frac{E}{2}\right)$ and radius

$$r = \frac{1}{2}\sqrt{D^2 + E^2 - 4C}.$$

Note the following

(a) If $D^2 + E^2 - 4C = 0$, then the equation represents a point which coincides with the center

$$C\left(-\frac{D}{2}, -\frac{E}{2}\right).$$

(b) If $D^2 + E^2 - 4C < 0$, then the radius is imaginary. The circle is referred to as a virtual circle.

(c) If $D^2 + E^2 - 4C > 0$, then the circle is real.

Example 3.2

Find the center and radius of the circle $x^2 + y^2 - 4x + 10y - 2 = 0$.

Solution

From the equation $D = -4, E = 10$ and $C = -2$. Hence the center is $C(2, -5)$. The radius is

$$\begin{aligned} r &= \frac{1}{2} \sqrt{D^2 + E^2 - 4C} \\ &= \frac{1}{2} \sqrt{16 + 100 + 8} \\ &= \sqrt{31} \end{aligned}$$

Exercise 3.1

1. Find the equation of the circle whose centre and radius are as given.

- (a) Center $(0,0)$, radius $r = 8$
- (b) Center $(-3,4)$, radius $r = 2$
- (c) Center $(-11,-3)$ radius $r = 5$
- (d) Center $\left(\frac{1}{3}, \frac{1}{2}\right)$ radius $r = \frac{2}{3}$

2. Find the center and the radius of the circle given by the equation and state whether real, a point or imaginary.

- (a) $x^2 + y^2 + 12x - 6y - 7 = 0$
- (b) $x^2 + y^2 + 2x + 3y + 5 = 0$
- (c) $x^2 + y^2 - 10x - 6y + 34 = 0$
- (d) $x^2 + 4x + 8y - 41 = 0$
- (e) $x^2 + y^2 - x - y - 3 = 0$

3. Find the equation of the circle through the given points.

- (a) $(1,2)$ $(3,7)$ $(2,-3)$
- (b) $(-7,3)$ $(1,1)$ $(-3,4)$
- (c) $(2,5)$ $(9,8)$ $(1,-10)$
- (d) $(-1,1)$ $(2,-3)$ $(7,-7)$
- (e) $(2,4)$ $(5,9)$ $(8,14)$

4. Sketch the graphs,

- (a) $x^2 + y^2 + 3x + 5y - 2 = 0$
- (b) $x^2 + y^2 - 8x - 6y + 3 = 0$
- (c) $4x^2 + 4y^2 + 2x - 10y - 1 = 0$
- (d) $x^2 + y^2 - 16 = 0$

3.2 Position of a point in relation to a circle

If point $P(a, b)$ lies on the circle $x^2 + y^2 = r^2$, then $a^2 + b^2 = r^2$. If it lies inside the circle, then $a^2 + b^2 < r^2$ and if it lies outside the circle then $a^2 + b^2 > r^2$. Similarly, if $P(a, b)$ lies on the circle $(x - h)^2 + (y - k)^2 = r^2$, then

$$(a - h)^2 + (b - k)^2 = r^2.$$

If the point is inside the circle, then

$$(a - h)^2 + (b - k)^2 < r^2$$

and if outside then

$$(a - h)^2 + (b - k)^2 > r^2.$$

Example 3.3

Determine whether the point $P(4, 5)$ lies on, inside or outside the circle $x^2 + y^2 = 35$.

Solution

Substitute the coordinates in the equation to get

$$4^2 + 5^2 = 31 < 35.$$

The point therefore lies inside the circle.

3.3 Distance from a point to a circle

Consider point $P_1(a_1, b_1)$ outside the circle $(x - h)^2 + (y - k)^2 = r^2$. Let B_1 be a point on the circle such that it lies on the line segment P_1C , C being the center of the circle.

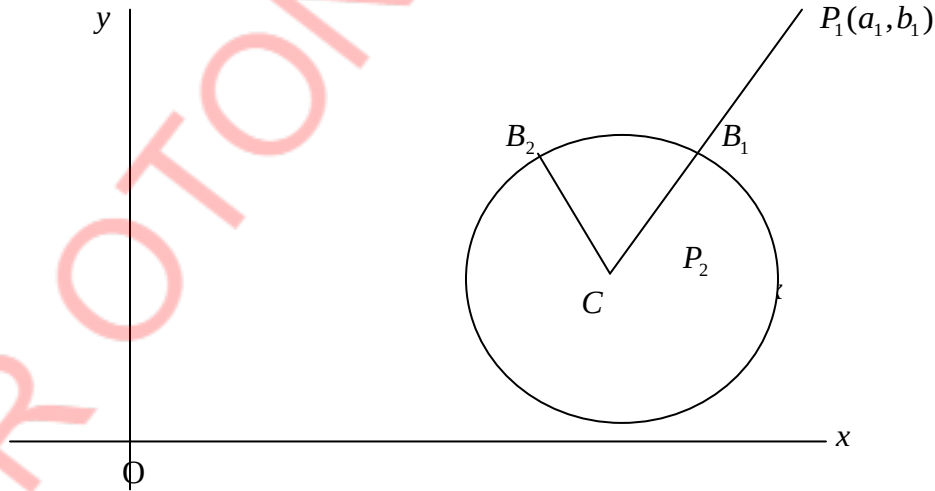


Figure 3.2

The distance from $P_1(a_1, b_1)$ to the circle is $\overline{P_1B_1}$ which is obtained as

$$\begin{aligned}\overline{P_1B_1} &= \overline{P_1C} - \overline{B_1C} \\ &= \sqrt{(a_1 - h)^2 + (b_1 - k)^2} - r\end{aligned}$$

Now consider point another $P_2(a_2, b_2)$ inside the circle lying on the line segment CB_2 . The distance from $P_2(a_2, b_2)$ to the circle is

$$\overline{P_2B_2} = r - \sqrt{(a_2 - h)^2 + (b_2 - k)^2}.$$

In general, the distance d from point $P(a, b)$ to circle $(x - h)^2 + (y - k)^2 = r^2$ is

$$d = \sqrt{(a - h)^2 + (b - k)^2} - r$$

with d positive if the point is outside the circle and negative if inside the circle.

Example 3.4

Find the distance from the point $P(5, 8)$ to the circle $x^2 + y^2 + 2x + 6y + 3 = 0$.

Solution

The center of the circle is $C(-1, -3)$. The radius of the circle is

$$\begin{aligned}r &= \frac{1}{2} \sqrt{D^2 + E^2 - 4C} \\ &= \frac{1}{2} \sqrt{16 + 36 - 12} \\ &= \sqrt{7}\end{aligned}$$

Hence the distance is

$$\begin{aligned}d &= \sqrt{(5 + 1)^2 + (8 + 3)^2} - \sqrt{7} \\ &= 9.88\end{aligned}$$

Exercise 3.2

- Find the distance from the point to the circle and state whether the point is outside, inside or on the circle.

(a) (8, 7)	to	$x^2 + y^2 + 12x - 5y + 3 = 0$
(b) (4, -3)	to	$x^2 + y^2 - 6x + 6y + 5 = 0$
(c) (-7, 6)	to	$3x^2 + 3y^2 + 6x - 5y - 1 = 0$
(d) (1, 1)	to	$2x^2 + 2y^2 - 1 = 0$
(e) (-6, 10)	to	$x^2 + y^2 + 16x + 20y + 160 = 0$

2. Find the equation of the circle with the given center and radius.
 - (a) C (0,7) $r = 3$
 - (b) C (1,-8) $r = 10$
 - (c) C (-7,0) $r = 0$
 - (d) C $(\frac{1}{4}, \frac{1}{5})$ $r = \frac{9}{2}$

3. Find the center and radius of the circle whose equation is and state whether real or imaginary
 - (a) $x^2 + y^2 - 6x - 12y - 4 = 0$
 - (b) $x^2 + y^2 - 4x - 8y + 20 = 0$
 - (c) $x^2 + y^2 + 7x + 2y - 6 = 0$
 - (d) $4x^2 + 4y^2 - 10x + 8y - 12 = 0$

4. Find the equation of the circle through the given points.
 - a. (2,1) (3,2) (4,5)
 - b. (-3,2) (1,7) (-1,-3)
 - c. (0,2) (5,3) (2,-4)

5. Find the distance from the point A to the given circle and state whether it lies inside, on the circle or outside if,
 - (a) A (2,3), $x^2 + y^2 - 8x + 10y - 7 = 0$
 - (b) A (-17,12) $x^2 + y^2 - x - y - 1 = 0$
 - (c) A (-10,10) $x^2 + y^2 + 7y + 8 = 0$

6. A circle passes through points (2,3) and (4,5) and its center is on the line $2x + 3y = 5$. Find the equation of the circle.

3.4 Tangent, secant, chord and normal

Definitions

Consider a circle with C (h,k) and radius r. The line AB through the circle intersecting with the circle at the points A_1 and B_1 is called a **secant**. Any line segment joining the two points on a circle is called a **chord**. For instance A_1B_1 is a chord. The chord through the center is called a diameter of the circle.

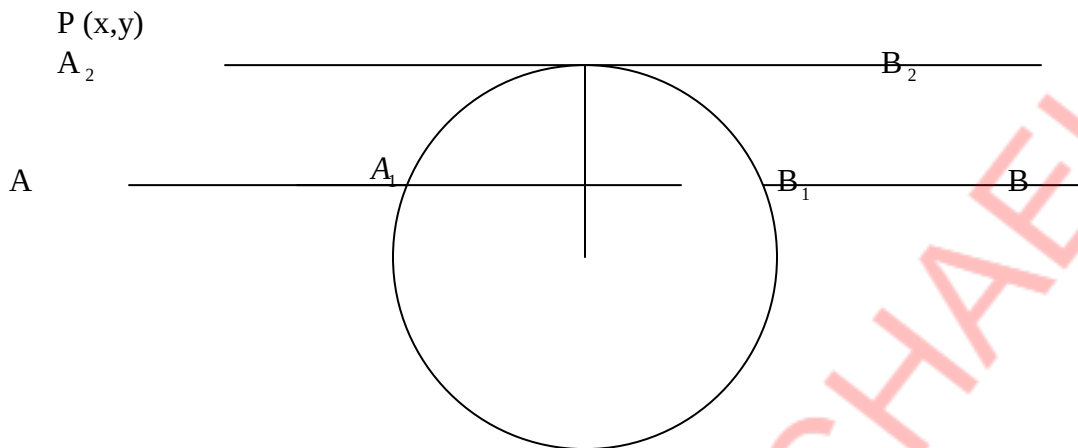


Figure 3.3

The line that has only a single point of contact with a circle is called the **tangent** to the circle at that point of contact. In the figure above A₂B₂ is a tangent to the circle at point P(x,y).

The **normal** to the circle at point P is the line through P which is perpendicular to the tangent at P. Example CP is a normal to the curve at P. The normal will always pass through the center of the circle.

3.4.1 Equation of the tangent

There are various methods of obtaining the equation of the tangent to a circle at a given point which may require application of differential calculus.

This is not considered, here we consider the following method. Consider the tangent AB

$P(x_1, y_1)$ at point P on the circle whose equation is $(x-h)^2 + (y-k)^2 = r^2$

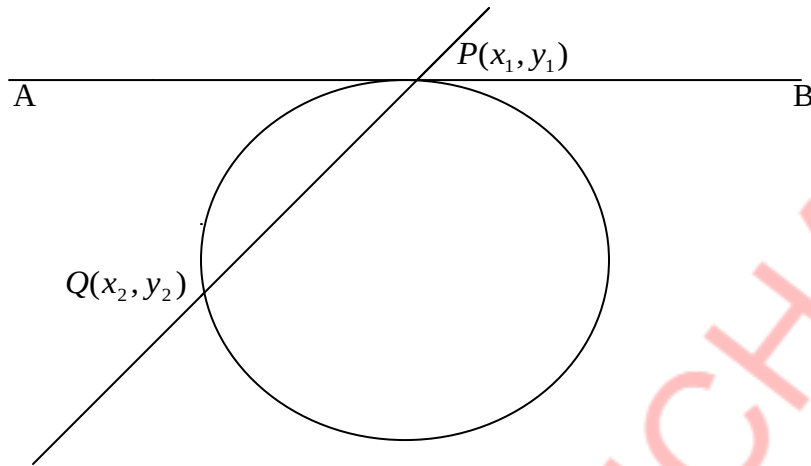


Figure 3.4

Let point $Q(x_2, y_2)$ on the circle move along the circle towards P. As Q approaches P, the line through P and Q approaches the tangent AP.

The equation of the line through P and Q is given by

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}, \quad y - y_1 = (x - x_1) \left(\frac{y_2 - y_1}{x_2 - x_1} \right)$$

the slope of this line, $m = \frac{y_2 - y_1}{x_2 - x_1}$ approaches that of AP is therefore the limit of m as Q approaches P which is given by

$$\lim_{Q \rightarrow P} \frac{y_2 - y_1}{x_2 - x_1} = \frac{0}{0}$$

But $\frac{0}{0}$ is undefined. To get the limit, consider that the two points Q and P lie on the circle and

$$\text{hence for P we have } (x_1 - h)^2 + (y_1 - k)^2 = r^2 \quad \text{--- (3.1) ---}$$

$$\text{For Q we have } (x_2 - h)^2 + (y_2 - k)^2 = r^2 \quad \text{--- (3.2) ---}$$

Subtracting (3.2) from (3.1)

$$\text{We get } (x_2 - h)^2 - (x_1 - h)^2 + (y_2 - k)^2 - (y_1 - k)^2 = 0$$

Or

$$(y_2 - k)^2 - (y_1 - k)^2 = (x_1 - h)^2 - (x_2 - h)^2$$

Or

$$[(y_2 - k) - (y_1 - k)][(y_2 - k) + (y_1 - k)] = [(x_1 - h) - (x_2 - h)][(x_1 - h) + (x_2 - h)]$$

Or

$$(y_2 - y_1)(y_1 + y_2 - 2k) = (x_1 - x_2)(x_1 + x_2 - 2h)$$

Or

$$\frac{y_2 - y_1}{x_2 - x_1} = -\frac{(x_1 + x_2 - 2h)}{(y_1 + y_2 - 2k)}$$

taking the limits gives

$$\lim_{Q \rightarrow P} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) = -\frac{(2x_1 - 2h)}{2y_1 - 2k} = -\frac{(x_1 - h)}{(y_1 - k)}$$

Hence the slope of the tangent at $P(x_1, y_1)$ is $-\frac{(x_1 - h)}{(y_1 - k)}$

Let $P(x, y)$ be any other point on the tangent. The equation of the tangent is

$$\frac{y - y_1}{x - x_1} = \frac{(x_1 - h)}{(y_1 - k)}$$

Or

$$y(y_1 - k) - y_1(y_1 - k) + x(x_1 - h) - x_1(x_1 - h) = 0$$

Or

$$xx_1 + yy_1 - hx - ky + hx_1 + ky_1 - x_1^2 - y_1^2 = 0$$

But

$$x_1^2 + y_1^2 = r^2 + 2hx_1 + 2ky_1 - h^2 - k^2 \quad \text{From 2.0} \quad \text{hence the equation is}$$

$$xx_1 + yy_1 - hx - ky - hx_1 - ky_1 + h^2 + k^2 - r^2 = 0 \quad \text{--- (3.3) ---}$$

Or

$$xx_1 + yy_1 - h(x + x_1) - k(y + y_1) + C = 0 \quad \text{--- (3.4) ---}$$

Note that equation 3.3 can be written as

$$(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2 \quad \text{--- (3.5) ---}$$

Example 3.5

Find the tangent to the circle at the point $P(7, 12)$ if the equation has center $C(4, 8)$

Solution

To get the tangent, first get the equation of the circle as follows. The radius of the circle is PC given by

$$r = PC = \sqrt{(4 - 7)^2 + (8 - 12)^2} = 5$$

Hence the equation of the circle is

$$(x - 4)^2 + (y - 8)^2 = 5$$

from 3.5 we get

$$(x-4)(7-4) + (y-8)(12-8) = 5$$

Or

$$3x - 12 + 4y - 32 = 5$$

Or

$$3x + 4y = 49$$

Example 3.6

Find the equation of the tangent to the circle $x^2 + y^2 + 10 - 6y - 66 = 0$ at the point (3,-3)

Solution

The equation is obtained as follows:-

The center of the circle is C(-5,3) Hence

$$3x - 3y + 5(x+3) - 3(y-3) - 66 = 0$$

Or

$$8x - 6y - 42 = 0$$

The equation of the tangent to a circle through a given point can also be obtained if the center of the circle is known. The tangent will be perpendicular to the normal which passes through the given point on the circle.

Example 3.7

Find the equation of the tangent to the circle with center C(2,5) at point P(6,3) on the circle.

Solution

The slope of the normal that passes through P(6,3) is

$$m = \frac{3-5}{6-2} = -\frac{1}{2}.$$

The slope of the tangent at P is therefore

$$m_1 = -\frac{1}{m} = 2.$$

Let R(x,y) be any other point on the tangent. Then

$$\frac{y-3}{x-6} = 2.$$

Which reduces to

$$2x - y - 9 = 0.$$

3.4.2 Tangent to a circle from a given point

Let point $P_1(x_1, y_1)$ be a point outside the circle $(x-h)^2 + (y-k)^2 = r^2$. Let point D(m,n) be the point on the circle such that the line through P_1 and D is a tangent to the circle. The problem is to determine the coordinates of D, m and n.

Since $D(m, n)$ is on the circle, the equation of the tangent is

$$(m-h)(x_1-h) + (n-k)(y_1-k) = r^2$$

Or

$$m(x_1-h) + n(y_1-k) = r^2 + h(x_1-h) + k(y_1-k)$$

Or

$$ma + nb - q = r^2 + h(x_1-h) + k(y_1-k) \quad \text{--- 3.6 ---}$$

This is a single equation with two unknowns and hence another equation in m and n is required to be able to go for m and n .

Since $D(m, n)$ is on the circle, we have

$$(m-h)^2 + (n-k)^2 = r^2 \quad \text{--- 3.7 ---}$$

Put $n = \frac{q-ma}{b}$

Substituting in 3.7 we get

$$(m-h)^2 + \left(\frac{q-ma}{b} - k\right)^2 = r^2$$

This is a quadratic equation in m and applying the usual solution techniques of quadratic equations, there will be two possible values of m say m_1 and m_2 . This implies there will be two possible values of n as n_1 and n_2 . Hence there are two points say $D_1(m_1, n_1)$ and $D_2(m_2, n_2)$ on the circle to which from $P_1(x_1, y_1)$ to the circle can be drawn.

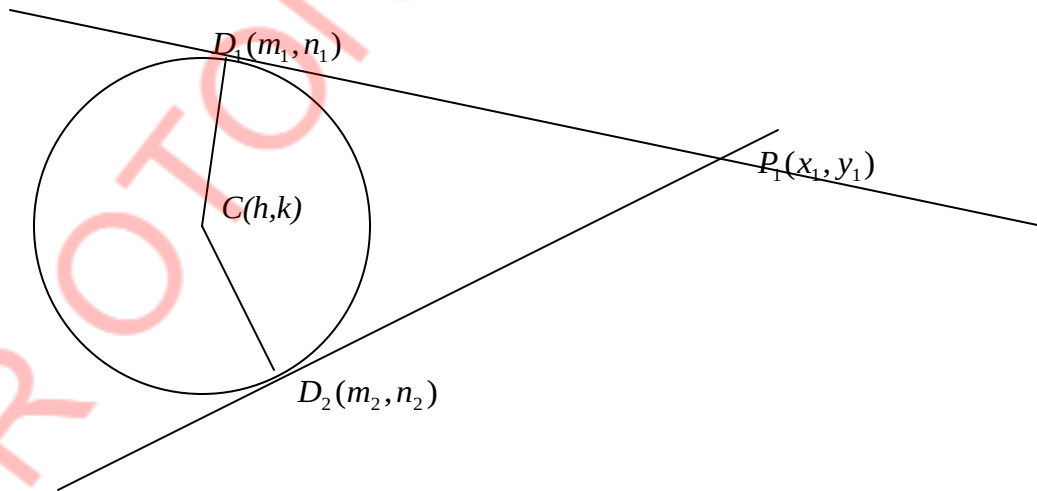


Figure 3.5

Remark

The chord through D_1 and D_2 is called the chord of contact of tangents from the point $P_1(x_1, y_1)$ with respect to the given circle.

Example 3.7

Find the tangent points of the circle $x^2 + y^2 = 5$ from the point $(8,3)$

Solution

Let (m, n) be the points on the circle and on the tangent

Then

$$\begin{aligned}
 mx + ny &= 5 & \Rightarrow 8m + 3n &= 5 & \text{and } m^2 + n^2 &= 5 \\
 \text{put } n &= \frac{5-8m}{3} \text{ then } m^2 + \frac{(5-8m)^2}{9} = 5 \\
 &\text{or} \\
 &9m^2 + 64m^2 - 80m + 25 = 45 \\
 &\text{or} \\
 &73m^2 - 80m - 20 = 0 \\
 \Rightarrow m &= \frac{80 \pm \sqrt{80^2 + 80 \times 73}}{146} = 1.33 \text{ or } -0.21
 \end{aligned}$$

$$\text{If } m = 1.33, n = \frac{5 - 8 \times 1.33}{3} = -1.88$$

$$\text{If } m = -0.21, n = \frac{5 - 8 \times -0.21}{3} = 2.23$$

Hence the two points on the circle are

$$D_1(1.33, -1.88) \text{ and } D_2(-0.21, 2.23)$$

Remark

The equation of the chord of contact can be easily obtained the moment D_1 and D_2 are obtained.

Example 3.8

Find the points on the circle $x^2 + y^2 + 12x + 4y + 15 = 0$ through which the tangents from the point $P_1(1, -3)$ pass. Also find the equation of the chord of contact.

Solution

Let $D(m, n)$ be a point on the circle. The equation of the circle can be written as

$(x+6)^2 + (y+2)^2 = 25$. The tangent through $P_1(1, -3)$ and through $D(m, n)$ is

$$(m+6)(1+6) + (n+2)(-3+2) = 25$$

$\Rightarrow$

$$7m - n = -15 \Rightarrow n = 7m + 15$$

Since $D(m, n)$ is on the circle, then

$$(m+6)^2 + (n+2)^2 = 25$$

or

$$(m+6)^2 + (7m+17)^2 = 25.$$

Solving for m gives $m=-2 \Rightarrow n=1$, or $m=-3 \Rightarrow n=-6$. Hence the two points are $D_1(-3, -6)$ and $D_2(-2, 1)$. The equation of the chord through the two points is $7x - y + 15 = 0$.

3. 5 Intersection of two circles.

Consider two circles with respective general equations $x^2 + y^2 + D_1x + E_1y + C_1 = 0$ and $x^2 + y^2 + D_2x + E_2y + C_2 = 0$. At the points of intersection both equations are satisfied and therefore

$$x^2 + y^2 + D_1x + E_1y + C_1 = x^2 + y^2 + D_2x + E_2y + C_2.$$

This gives

$$(D_1 - D_2)x + (E_1 - E_2)y + (C_1 - C_2) = 0,$$

which is the equation of a line. The points of intersection lie on this line. Making y the subject and substituting in any of the circle equations and solving will give the coordinates of the points.

Example 3.9

Find the points of intersection of the circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 - 8x - 2y - 19 = 0$. Equating the two equations gives the linear equation $4x - 4y + 7 = 0$ or $y = x + 7/4$. Substituting in the first equation we get the quadratic equation $32x^2 - 104x - 311 = 0$. Hence

$$x = \frac{104 \pm \sqrt{50624}}{64}.$$

This finally gives the points of intersection as $()$ and $()$.

3.6 Orthogonal circles

Consider two circles, which intersect. The circles are said to be orthogonal if their respective tangents at the point of intersection are perpendicular to each other. From the diagram below,

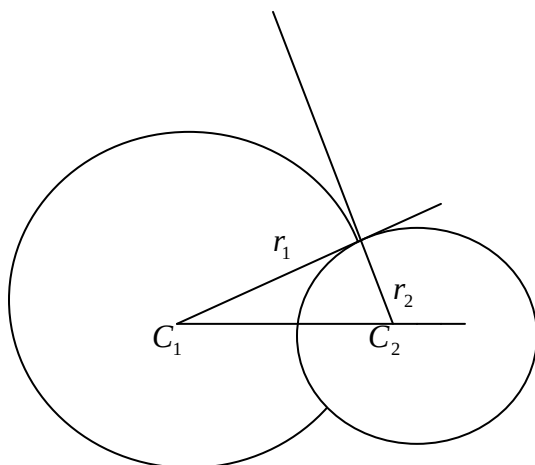


Figure 3.6

$\angle C_1PC_2$ is right angled and $C_1C_2^2 = r_1^2 + r_2^2$

Hence if two circles are orthogonal, the square of the distance between their centers is equal to the sum of squares of their radii.

Example 3.10

Show that the circles $x^2 + y^2 + 6x - 4y - 12 = 0$ and $x^2 + y^2 - 20x - 4y - 40 = 0$ are orthogonal.

Solution

The center of the first circle is $(-3, 2)$ and its radius is $r_1 = \sqrt{36 + 16 + 48} = 10$

The center of the second circle is $(10, 2)$ and radius is $r_2 = \frac{1}{2}\sqrt{400 + 16 + 160} = 12$

The distance between the centers is $d = \sqrt{13^2 + 0} = 13$

Also $r_1^2 + r_2^2 = 10^2 + 12^2 = 13^2$, hence they are orthogonal.

Exercise 3.3

1. Find the equation of the circle whose diameter is the line joining the points given.

- (a) $(2, 2)$ $(7, 9)$
- (b) $(6, 8)$ $(12, 1)$
- (c) $(-3, 1)$ $(-1, -1)$
- (d) $(8, 4)$ $(2, 9)$
- (e) $(7, 10)$ $(21, 3)$

2. Find the equation of the normal to the given circle at the given point.

- (a) $x^2 + y^2 + 4x - 5y - 2 = 0$ $(1, \sqrt{2})$
- (b) $3x^2 + 3y^2 - 2x - 16y - 52 = 0$ $(4, 6)$

- (c) $x^2 + y^2 - 2x + 8y - 23 = 0$ (3,-10)
 (d) $x^2 + y^2 - 4x + 6y - 37 = 0$ (3,4)
3. Find the equation of the tangents to the given circle from the given external point.
- (a) $x^2 + y^2 - 10 = 0$ (10,2)
 (b) $x^2 + y^2 + 4x - 4y - 8 = 0$ (3,2)
 (c) $3x^2 + 3y^2 + 2x + 4y - 6 = 0$ (5,-3)
 (d) $x^2 + y^2 - 5x + 10y - 12 = 0$ (-9,-6)
4. Find the equation of the normal to the given circle at the given point.
- (a) $x^2 + y^2 - 18x - 16y + 120 = 0$ (6,4)
 (b) $x^2 + y^2 - 6x - 10y + 9 = 0$ (1,8)
 (c) $x^2 + y^2 - 2x + 14y - 119 = 0$ (6,5)
 (d) $x^2 + y^2 + 12x - 4y - 60 = 0$ (2,-4)

Miscellaneous Exercises

1. Find the standard equation of the circle whose center and radius are as given.
- (a) Center (0,0), radius $r=10$
 (b) Center (6,5), radius $r=3$
 (c) Center (-2,0), radius $r=8$
 (d) Center (-13,13), radius $r=13$
2. Find the general equation of the circle whose center and radius are as given.
- (a) Center (1, -9), radius $r=4$
 (b) Center (-8,0), radius $r=10$
 (c) Center (a, 12), radius $r=a-3$
 (d) Center (-17,17), radius $r=17$
3. Find the equation of the circle, which passes through the given points.
- (a) A (1,2) B (3,2), C (5,2)
 (b) A (-4, 0) B (8, 12) C (1, -1)
 (c) A (10,-18) B(7,-3) C(2,-2)
 (d) A (7,0) B (0,7) C(-2,0)
4. Find the center and the radius of the circle and state whether real or virtual.
- (a) $x^2 + y^2 = 30$
 (b) $x^2 + y^2 - 8x + 12y + 3 = 0$
 (c) $x^2 + y^2 + x + y + 1 = 0$
 (d) $4x^2 + 4y^2 - 8x + 3y - 1 = 0$
5. Determine whether the given point lies inside, outside or within the circle.
- (a) P(2,2), $x^2 + y^2 = 9$
 (b) P(4,5) $x^2 + y^2 - 12x - 10y + 36 = 0$

- (c) $P(-12,1)$ $x^2 + y^2 + 20x + 10y - 7 = 0$
 (d) $P(-3,7)$ $x^2 + y^2 - 12x + 25y - 8 = 0$
6. Find the distance from the given point to the circle.
 (a) $P(1,1)$ $x^2 + y^2 - 20x - 18y + 180 = 0$
 (b) $P(8,11)$ $x^2 + y^2 - 7 = 0$
 (c) $P(-2,-6)$ $x^2 + y^2 - 16x + 10y + 40 = 0$
 (d) $P(1,8)$ $4x^2 + 4y^2 - 4x + 8y + 5 = 0$
7. Find the equation of the tangent on the given circle at the given point on the circle.
 (a) $x^2 + y^2 + 2x - 6y - 3 = 0$, $P(2,5)$.
 (b) $x^2 + y^2 - 10x + 4y + 24 = 0$, $P(3,-3)$.
 (c) $x^2 + y^2 - 12x + 5y + 13 = 0$, $P(6+\sqrt{23},0)$
 (d) $x^2 + y^2 - 4x + 6y - 37 = 0$, $P(3,4)$
8. Find the points on the circle whose tangents to the circle intersect at the given point in question 6 above.
9. Find the points of intersection of the circles given if they intersect.
 (a) $x^2 + y^2 + 4x - 6y - 13 = 0$, and $x^2 + y^2 - 25 = 0$,
 (b) $x^2 + y^2 + 7x - 8y - 36 = 0$, and $x^2 + y^2 - 24x + 26y - 312 = 0$,

Summary

In this chapter we have discussed the circle where we have shown how to:

- Determine the general and standard equation of a circle.
- Find the tangent and the normal to a circle at a given point on the circle.
- Find the tangents to a circle from a given external point.
- Find points of intersection of two circles.
- Find the distance from a point to a circle.

Further Reading

- L. Bostock, S. Chandler, and C. Rourke. **Further Pure Mathematics**. Stanley Thornes (Publishers) Ltd, 1982.
- R.E Larson and R.P. Hostetler. **Precalculus**. Houghton Mifflin Company, 1997.
- P.K. Jain and Khalil Ahmad. **Analytical Geometry of Two Dimensions**. Wiley Eastern Limited, 1983.

Chapter 4: PARABOLA

4.0 Introduction

In the next three chapters we study the conic sections, the parabola, the ellipse and the hyperbola. A conic section is the locus of a point described such that the ratio of its distance from a fixed point to that from a fixed line is a constant. This ratio is called the eccentricity, denoted by e . The fixed line is called the directrix and the fixed point the focus.

Definition: A parabola is the locus of a point which moves such that the ratio of its distance from a fixed point to the distance from a fixed line is one.

Objectives

At the end of the chapter the students should be able to:-

- (a) Find the standard and general equation of a parabola given its focus and directrix.
- (b) Find the tangent and the normal to the parabola at a given point on the parabola.
- (c) Find the parametric equation of a parabola.

4.1 Standard equation of a parabola

Let the focus be denoted by F and the directrix be DM . The line passing through the focus and perpendicular to the directrix is called the axis of the parabola (Fig 1). The point V where the parabola intersects with the axis is called its vertex.

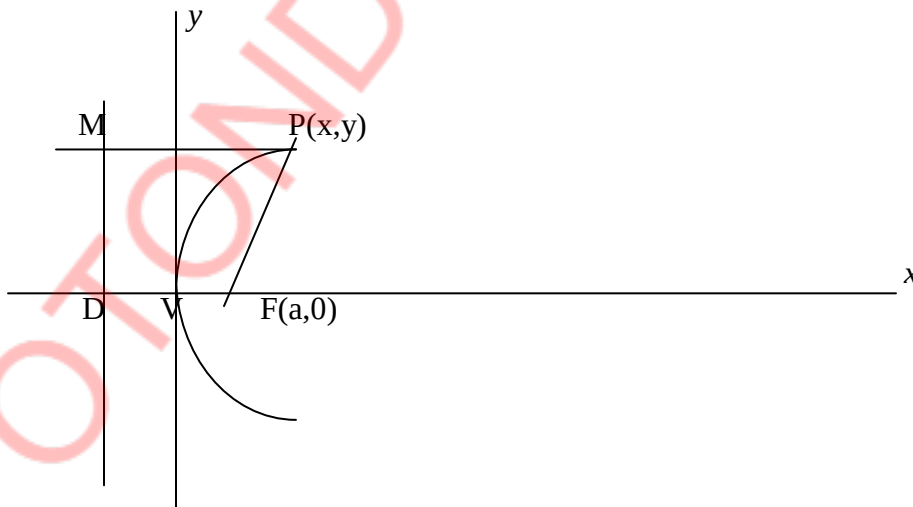


Figure 4.1

To get the equation of the parabola, suppose the axis coincides with the x-axis and let the vertex V be at the origin. Let the distance from the vertex to the focus be a . The focus is $F(a,0)$. From the definition of the parabola,

$$\frac{PF}{MP} = 1 ,$$

where $P(x, y)$ is any other point on the parabola. The distances PF and PM are

$$PF = \sqrt{(x-a)^2 + y^2} \quad \text{and} \quad PM = x + a$$

Thus

$$\sqrt{(x-a)^2 + y^2} = x + a$$

which gives

$$(x-a)^2 + y^2 = (x+a)^2$$

or

$$x^2 - 2ax + a^2 + y^2 = x^2 + 2ax + a^2$$

or

$$y^2 = 4ax$$

This is the standard equation of a parabola whose axis is horizontal and the vertex is at the origin. A chord (a line segment whose end points are points on the parabola) passing through the focus F is referred to as the focal chord. The focal chord perpendicular to the axis is called the latus rectum.

The distance of any point $P(x, y)$ on the curve to the focus is referred to as its focal distance. The length of the latus rectum is equal to $4a$.

Example 4.1

Find the equation of the parabola whose focus is $F(2, 0)$ and the directrix is $x = -2$.

Solution

Since the focus is $F(2, 0)$ and the directrix is $x = -2$, then the vertex is at the origin and its axis is the x -axis. Hence $a = 2$. The equation is

$$y^2 = 4ax = 8x$$

Example 4.2

Find the focus, the equation of the directrix, the length of latus rectum for the parabola

$$5y^2 = 24x.$$

Solution

Write the equation in standard form to get

$$y^2 = \frac{24}{5}x.$$

This implies

$$4a = \frac{24}{5} \Rightarrow a = \frac{6}{5}.$$

Hence the focus is $F(6/5, 0)$.

The equation of the directrix is

$$x = -a = -\frac{6}{5}.$$

The length of the latus rectum is

$$l = 4a = \frac{24}{5}$$

4.2 General equation of a parabola

Let the focus of the parabola be $F(h,k)$ and the equation of the directrix be $Ax'+By'+C=0$ (Fig.2)

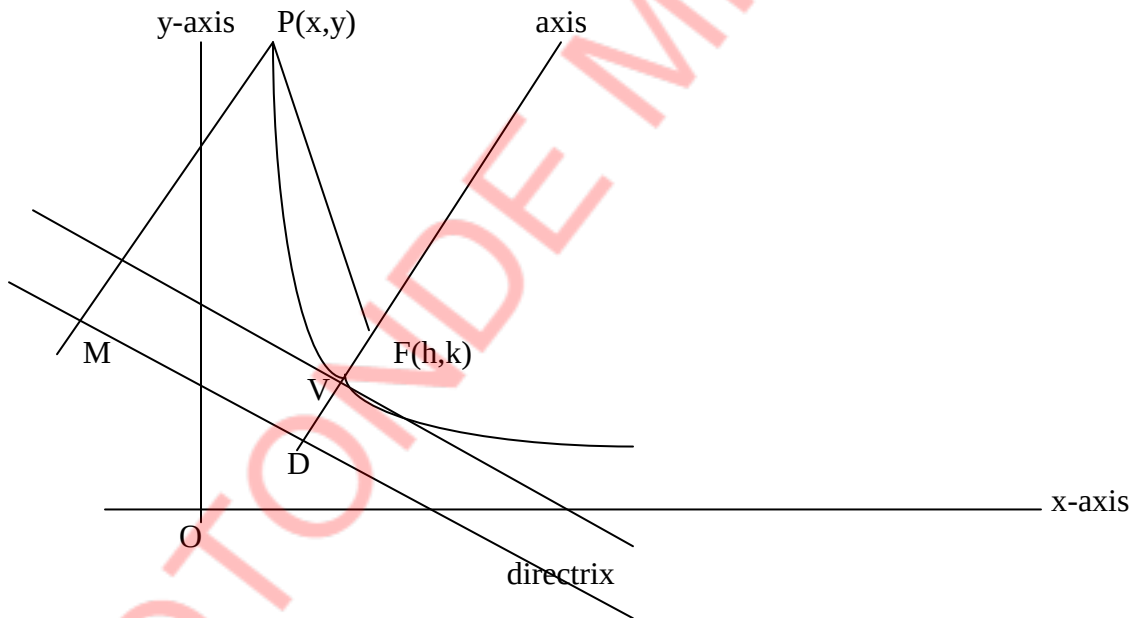


Figure 4.2

Let $P(x,y)$ be any point on the parabola. From the definition of parabola, $FP=PM$.

Also

$$FP = \sqrt{(x-h)^2 + (y-k)^2}.$$

The distance from point (x,y) to line $Ax+By+C=0$ is $MP = \frac{(Ax + By + C)}{\sqrt{A^2 + B^2}}$

The general equation of the parabola is therefore given by

$$(x-h)^2 + (y-k)^2 = \frac{(Ax + By + C)^2}{A^2 + B^2}$$

Special cases

(a)

From the general equation, the equation of a parabola whose axis is y-axis focus $F(0,a)$ and directrix is $y = -a$ or $y+a=0$ (hence vertex is at origin) is given by

$$x^2 + (y-a)^2 = (y+a)^2$$

$\Rightarrow$

$$x^2 = 4ay$$

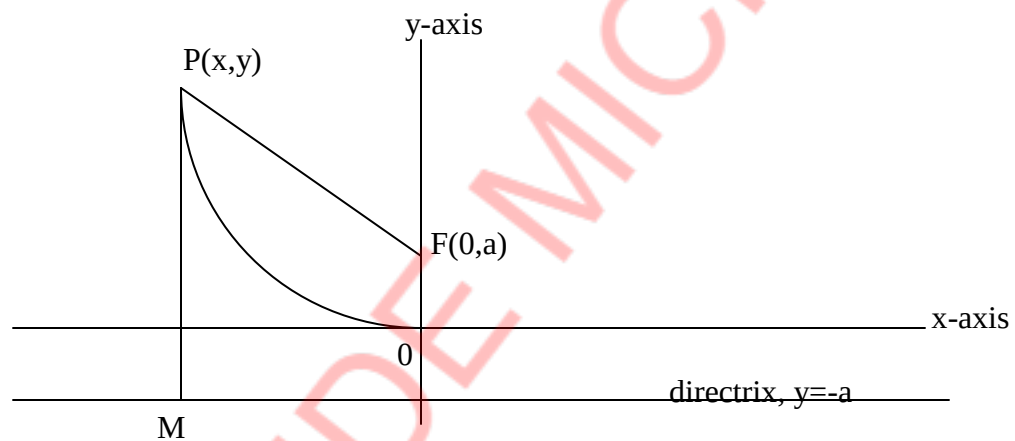


Figure 4.3

(b) The vertex at the origin, focus is $F(-a,0)$ and directrix $x=a$ or $x-a=0$. The equation is

$$(x+a)^2 + y^2 = (x-a)^2$$

$\Rightarrow$

$$y^2 = -4ax$$

BY ANASTASIA NDUTA

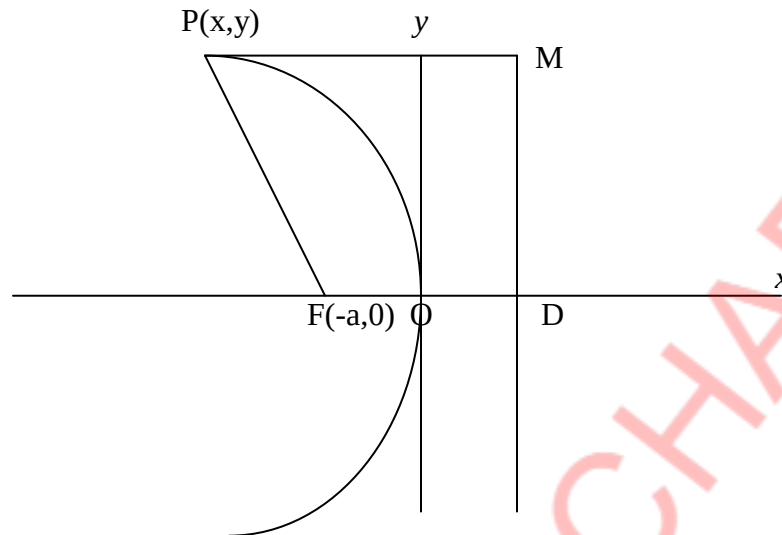


Figure 4.4

(c)

The vertex at the origin, focus is $F(0,-a)$ and directrix $y=a$ or $y-a=0$. The equation is

$$x^2 + (y-a)^2 = (y-a)^2$$

$\Rightarrow$

$$x^2 = -4ay$$

y-axis

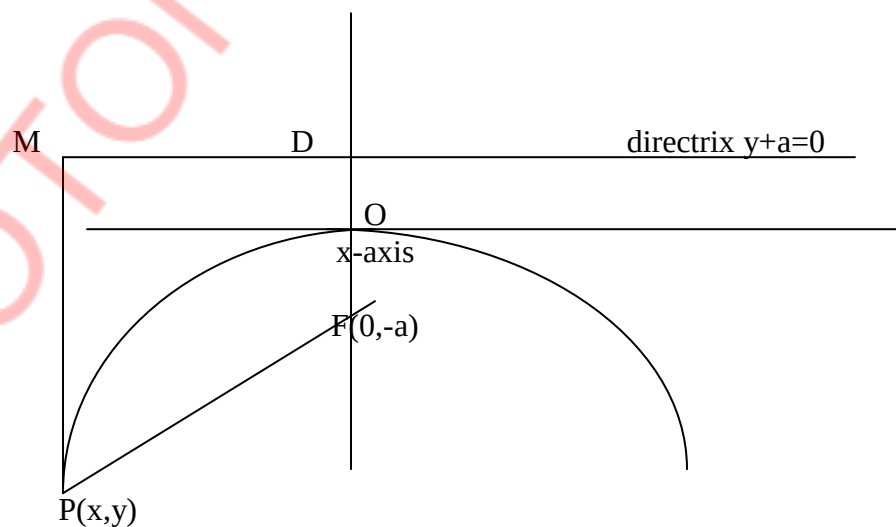


Figure 4.5

4.3 Parabola with horizontal or vertical axis but vertex not at origin

Consider a parabola with a horizontal axis, directrix $x=d$, focus $F(h,k)$ and vertex $V(c_1, c_2)$.

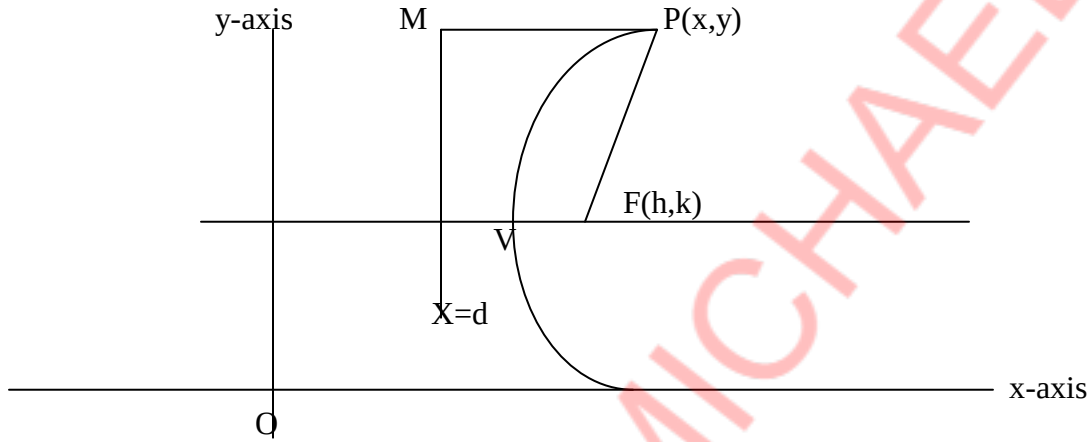


Figure 4.6

From the definition

$$(x-d)^2 = (x-h)^2 + (y-k)^2.$$

Let the distance from the vertex to the focus be a . Then $d = c_1 - a$, $h = c_1 + a$, and $k = c_2$. Therefore

$$(x - c_1 + a)^2 = (x - c_1 - a)^2 + (y - c_2)^2.$$

Which reduces to

$$(y - c_2)^2 = 4a(x - c_1)$$

It can be easily shown that if the axis is vertical, focus and vertex being $F(h,k)$ and $V(c_1, c_2)$, and directrix $y= d$, then the equation of the parabola is

$$(x - c_1)^2 = 4a(y - c_2)$$

Example 4.3

Find the vertex, the axis of symmetry, the focus, the equation of the directrix and the length of the latus rectum of the parabola

$$y^2 - 6y - 12x + 9 = 0$$

Solution

Write the equation in standard form as

$$y^2 - 6y + 9 = 12x$$

$\Rightarrow$

$$(y-3)^2 = 12x$$

This implies $4a = 12 \Rightarrow a = 3$. Hence the vertex is $V(0,3)$, the axis of symmetry is $y=3$, the equation of the directrix is $x = d = c_1 - a = -3$, that is

$$x = -3.$$

Example 4.4

Find the vertex, the axis of symmetry, the focus, the equation of the directrix and the length of the latus rectum of the parabola

$$x^2 - 10x - 6y + 40 = 0$$

Solution

Write the equation in standard form as

$$x^2 - 10x + 40 = 6y.$$

This reduces

$$(x-5)^2 = 6(y-5/2)$$

This implies $4a = 6 \Rightarrow a = 3/2$. Hence the vertex is $V(5,5/2)$, the axis of symmetry is $x=5$, the equation of the directrix is $x = d = c_1 - a = 4$, that is

$$x = 4.$$

Exercises 4.1

- Find the equation of the hyperbola whose focus and directrix are as given below.
 - $F(5,0)$, $x+5=0$
 - $F(0,-2)$, $y-2=0$
 - $F(3,4)$, $4x+3y-2=0$
 - $F(6,-2)$, $3y=8x-4$
- Find the vertex, the focus, the equation of the directrix and the length of the latus rectum.
 - $9x^2 + 24x - 252y + 16 = 0$
 - $x^2 + 8x - 2y + 10 = 0$
 - $4y^2 - 4y + 3x + 193 = 0$
 - $98y^2 - 56y - 84x + 176 = 0$

4.4 Tangent to a parabola

We consider the case when the axis is horizontal and when it is vertical.

For the horizontal case, let the equation of the parabola be $(y-c_2)^2 = 4a(x-c_1)$. To get the equation of the tangent to the parabola at point $P(x_1, y_1)$, consider another point $Q(x_2, y_2)$ (Fig 4.7)

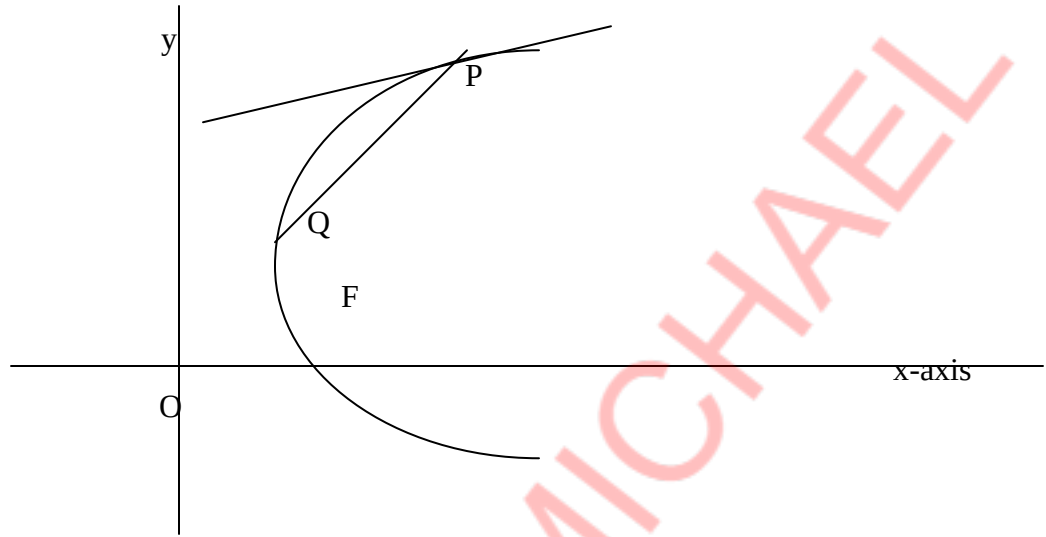


Figure 4.7

The slope of PQ is

$$m_{PQ} = \frac{y_2 - y_1}{x_2 - x_1}.$$

As Q approaches P, x_2 approaches x_1 and y_2 approaches y_1 . The limit of the slope of PQ is therefore

$$\lim_{Q \rightarrow P} (m_{PQ}) = \lim_{Q \rightarrow P} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) = \frac{0}{0}$$

which is undefined. To get the limit, consider the equations

$$(y_2 - c_2)^2 = 4a(x_2 - c_2) \text{ -----(4.1)}$$

$$(y_1 - c_2)^2 = 4a(x_1 - c_1) \text{ -----(4.2)}$$

Subtracting (4.2) from (4.1) gives

$$(y_2 - c_2)^2 - (y_1 - c_2)^2 = 4a(x_2 - x_1)$$

or

$$[(y_2 - c_2) - (y_1 - c_2)][(y_2 - c_2) + (y_1 - c_2)] = 4a(x_2 - x_1)$$

$\Rightarrow$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{4a}{y_1 + y_2 - 2c_2}$$

Hence as Q approaches P the limit of the slope is

$$m_t = \lim_{Q \rightarrow P} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) = \frac{2a}{y_1 - c_2}.$$

Let R(x,y) be any other point on the tangent. Then the equation of the tangent is

$$\frac{y - y_1}{x - x_1} = \frac{2a}{y_1 - c_2}$$

or

$$(y - y_1)(y_1 - c_2) = 2a(x - x_1) .$$

This can be restructured as below.

$$[(y - c_2) - (y_1 - c_2)](y_1 - c_2) = 2a[(x - c_1) - (x_1 - c_1)]$$

$\Rightarrow$

$$(y - c_2)(y_1 - c_2) = (y_1 - c_2)^2 + 2a(x - c_1) - 2a(x_1 - c_1)$$

$\Rightarrow$

$$(y - c_2)(y_1 - c_2) = 4a(x_1 - c_1) + 2a(x - c_1) - 2a(x_1 - c_1)$$

$\Rightarrow$

$$(y - c_2)(y_1 - c_2) = 2a(x + x_1 - 2c_1) .$$

When the vertex is at the origin, the equation of the tangent at point $P(x_1, y_1)$ is

$$yy_1 = 2a(x + x_1) .$$

Example 4.5

Find the equation of the tangent to the parabola $y^2 - 2y - 12x - 23 = 0$ at point $P(-7/4, 10)$.

Solution

Write the equation in standard form as

$$y^2 - 2y + 1 = 12x + 24$$

$$(y - 1)^2 = (x + 2)$$

Hence $c_1 = -2, c_2 = 1$ and $a = 3$. The equation of the tangent at point $P(-7/4, 10)$ is therefore

$$(y - 1)(10 - 1) = 2 \times 3(x - 7/4 - 4)$$

or

$$6x - 6y - 17 = 0 .$$

Example 4.6

Find the equations of the tangents at the end points of the latus rectum of the parabola

$$y^2 + 6y - 20x + 49 = 0 .$$

Solution

Write the equation in standard form to get

$$(y + 3)^2 = 20(x - 2) .$$

This implies $4a = 20, \Rightarrow a = 5$. The axis of symmetry is $y = -3$. The vertex is $V(2, -3)$, the focus is $F(7, -3)$ and the endpoints of the latus rectum are $P_1(7, 7)$ and $P_2(7, -13)$.

The equation of the tangent at $P_1(7, 7)$ is

$$(y + 3)(7 + 3) = 10(x + 7 - 14)$$

or

$$x - y - 10 = 0$$

The equation of the tangent at $P_2(7, -13)$ is

$$(y + 3)(-13 + 3) = 10(x + 7 - 14)$$

or

$$x + y - 4 = 0.$$

Now consider the case when the axis is vertical with the equation $(x - c_1)^2 = 4a(y - c_2)$.

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be points on the parabola (Fig below).

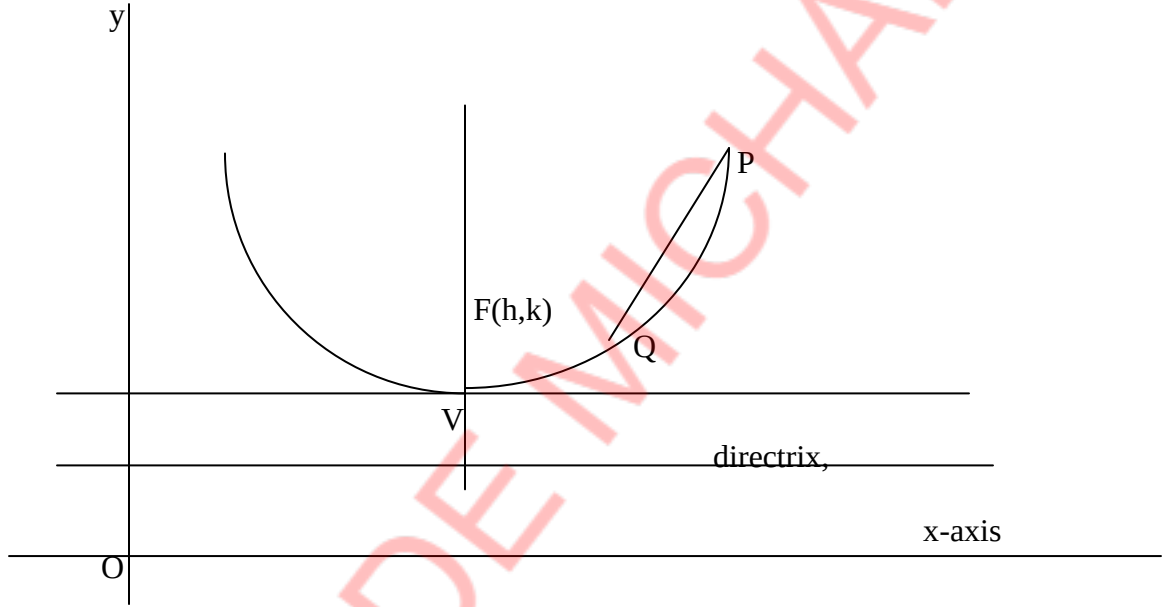


Figure 4.8

To get the slope of the tangent at $P(x_1, y_1)$ consider the slope of PQ which again is

$$m_{PQ} = \frac{y_2 - y_1}{x_2 - x_1}.$$

The limit of the slope as Q approaches P is

$$\lim_{Q \rightarrow P} (m_{PQ}) = \lim_{Q \rightarrow P} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) = \frac{0}{0}$$

which again is undefined. The coordinates of P and Q satisfy the equations

$$(x_2 - c_1)^2 = 4a(y_2 - c_2) \text{-----(4.3)}$$

$$(x_1 - c_1)^2 = 4a(y_1 - c_2) \text{-----(4.4)}$$

Subtracting (4.4) from (4.3) gives

$$(x_2 - c_1)^2 - (x_1 - c_1)^2 = 4a(y_2 - y_1)$$

or

$$[(x_2 - c_1) - (x_1 - c_1)][(x_2 - c_1) + (x_1 - c_1)] = 4a(y_2 - y_1)$$

$\Rightarrow$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{x_1 + x_2 - 2c_1}{4a}$$

Hence as Q approaches P the limit of the slope is

$$m_t = \lim_{Q \rightarrow P} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) = \frac{x_1 - c_1}{2a}.$$

Let R(x,y) be any other point on the tangent. Then the equation of the tangent is

$$y - y_1 = \frac{x_1 - c_1}{2a}(x - x_1).$$

or

$$(x - x_1)(x_1 - c_1) = 2a(y - y_1).$$

On restructuring the equation we get

$$(x - c_1)(x_1 - c_1) = 2a(y + y_1 - 2c_2).$$

When the vertex is at the origin, the equation of the tangent at point $P(x_1, y_1)$ is

$$xx_1 = 2a(y + y_1).$$

Example 4.7

Find the equation of the tangent to the parabola $x^2 - 22x + 24y + 169 = 0$ at the point whose focal distance is 11 units.

Solution

Write the equation in standard form as

$$(x - 11)^2 = -24(y + 2).$$

This implies the axis is vertical, the vertex is $V(11, -2)$, $4a = -24 \Rightarrow a = 6$ and the focus is $F(11, -8)$. Let $P(x_1, y_1)$ be a point whose focal distance is 11 units. Then

$$(x_1 - 11)^2 + (y_1 + 8)^2 = 121$$

Also

$$(x_1 - 11)^2 = -24(y_1 + 2).$$

$\Rightarrow$

$$(y_1 + 8)^2 - 24(y_1 + 2) = 121.$$

$\Rightarrow$

$$y_1 = -7 \text{ or } 15.$$

But $y_1 \leq -2$. For $y_1 = -7$, $x_1 = 11 + \sqrt{120}$ or $x_1 = 11 - \sqrt{120}$. Points $(11 + \sqrt{120}, -7)$ and $(11 - \sqrt{120}, -7)$ are at a distance of 11 units from the focus.

The equation of the tangent to the parabola at $(11 + \sqrt{120}, -7)$ is

$$(x-11)(11+\sqrt{120}-11) = -2 * 6(y-7+4)$$

or

$$\sqrt{120}x + 24y = 36 + 11\sqrt{120}.$$

4.5 Normal to a parabola

The normal to a parabola at a given point on the parabola is the line perpendicular to the tangent at the point. The slope of the normal is therefore the negative reciprocal of the slope of the tangent. Suppose the equation of the parabola is $(y - c_2)^2 = 4a(x - c_1)$. The slope of the tangent at point $P(x_1, y_1)$ on the parabola was found to be

$$m_t = \frac{2a}{y_1 - c_2}.$$

The slope of the normal is therefore

$$m_n = \frac{-(y_1 - c_2)}{2a}.$$

Let $R(x, y)$ be any other point on the normal. The equation of the normal is then

$$y - y_1 = \frac{-(y_1 - c_2)}{2a}(x - x_1).$$

Example 4.8

Find the equations of the tangent and normal to the parabola

$$y^2 - 10y + 2x + 41 = 0$$

at the point $(-10, 3)$.

Solution

The standard form is

$$(y - 5)^2 = -2(x + 8).$$

This implies $4a = 2, \Rightarrow a = 1/2$. The axis of symmetry is $y=5$. The vertex is $V(-8, 5)$, the focus is $F(-5, 5)$. The slope of the tangent at $(-10, 3)$ is

$$m_t = \frac{-2a}{y_1 - 5} = 1/2.$$

The equation of the tangent is therefore

$$(y - 5)(3 - 5) = -2 * 1/2(x - 10 + 16)$$

or

$$x - 2y + 16 = 0.$$

The equation of the normal at $(-10, 3)$ is

$$y - 3 = 2(x + 10)$$

or

$$2x - y + 23 = 0.$$

4.6 Pair of tangents from a point

Consider the parabola $(y - c_2)^2 = 4a(x - c_1)$ and point $Q(q_1, q_2)$ outside the parabola. There are two tangents to the parabola which intersect at $Q(q_1, q_2)$. Let $P(x_1, y_1)$ be a point on the parabola such that the tangent at that point passes through $Q(q_1, q_2)$. From the equation of the tangent through $P(x_1, y_1)$ and any other point $R(x, y)$ on the tangent,

$$(y - c_2)(y_1 - c_2) = 2a(x + x_1 - 2c_1).$$

$\Rightarrow$

$$(q_2 - c_2)(y_1 - c_2) = 2a(q_1 + x_1 - 2c_1). \text{-----(4.5)}$$

Since $P(x_1, y_1)$ is on the parabola, then

$$(y_1 - c_2)^2 = 4a(x_1 - c_1) \text{-----(4.6)}$$

Solving for x_1 and y_1 by equations (4.5) and (4.6) gives the two points on the parabola whose tangents meet at $Q(q_1, q_2)$.

Example 4.9

Find the points on the parabola $y^2 - 12y - 14x + 162 = 0$ through which the tangents which meet at $Q(2, 4)$ pass.

Solution

The standard form of the equation is $(y - 6)^2 = 14(x - 9)$. Let $P(x_1, y_1)$ be a point on the parabola and on tangent through $Q(2, 4)$. Then

$$(4 - 6)(y_1 - 6) = 7(2 + x_1 - 18)$$

$\Rightarrow$

$$7x_1 + 2y_1 - 114 = 0$$

and

$$(y_1 - 6)^2 = 14(x_1 - 9).$$

Solving for x_1 and y_1 gives $y_1 = 4 \pm \sqrt{82} \Rightarrow x_1 = \frac{106 \pm 2\sqrt{82}}{7}$. Having obtained the coordinates of the points the equations of the tangents can be easily obtained.

4.7 Parametric equation of a parabola

Consider the parabola

$$(y - c_2)^2 = 4a(x - c_1).$$

Let

$$(y - c_2) = 2at.$$

Then

$$4a(x - c_1) = 4a^2t^2, \Rightarrow x - c_1 = at^2.$$

This implies the parametric equation of a parabola with its vertex at $C(c_1, c_2)$ is

$$x = at^2 + c_1, y = 2at + c_2.$$

The point $(at^2 + c_1, 2at + c_2)$ is referred to as point t of the parabola.

If the vertex of the parabola is at the origin, the parametric equation is

$$x = at^2, y = 2at.$$

And point t of the parabola is $(at^2, 2at)$.

Example 4.10

Find the parametric equation of the parabola $y^2 = 23x$.

Solution

The parabola has its center at the origin and $4a = 23, \Rightarrow a = 23/4$. The parametric equation is therefore $x = 23t^2 / 4, y = 23t / 2$

Example 4.11

Find the parametric equation of the parabola

$$y^2 - 12y - 20x + 76 = 0.$$

Solution

The equation can be written as $(y - 6)^2 = 20(x - 2)$. The vertex of the parabola is $V(2, 6)$ and $4a = 20, \Rightarrow a = 5$. Hence the parametric equation of the parabola is

$$x = 5t^2 + 2, y = 10t + 6.$$

The equation of the tangent to the parabola at point t_1 can be obtained as follows. In the equation of the tangent at point $P(x_1, y_1)$ on parabola $(y - c_2)^2 = 4a(x - c_1)$,

$$(y - c_2)(y_1 - c_2) = 2a(x + x_1 - 2c_1),$$

substitute $(y_1 - c_2)$ with $2at_1$, and $x_1 - c_1$ with at_1^2 to get

$$t_1(y - c_2) = (x + at_1^2 - c_1).$$

Example 4.12

Find the equation of the tangent at point $t = 2$ on the parabola $y^2 + 14y - 12x + 13 = 0$.

Solution

In standard form the equation is $(y + 7)^2 = 12(x + 3)$ which implies $c_1 = -3, c_2 = -7$,
 $4a = 12, \Rightarrow a = 3$. From (eq) the equation of the tangent at $t = 2$ is

$$2(y + 7) = (x + 3 \cdot 4 + 3)$$

which reduces to

$$x - 2y + 1 = 0.$$

This can be easily verified as follows. Substituting $t = 2$ in (eq) the Cartesian coordinates of the point are (9,5). The equation of the tangent through this point from (eq) is

$$y + 7)(5 + 7) = 6(x + 9 + 6), \Rightarrow x - 2y + 1 = 0.$$

Exercises 4.3

Find the equation of the tangent to the parabola at the given value of t .

- (a) $y^2 + 3y - 2x + 1 = 0$ at $t=4$
- (b) $y^2 - 20y + 9x + 8 = 0$ at $t=3$
- (c) $x^2 - 24x - 2y - 12 = 0$ at $t=5$

Miscellaneous Exercises

1. Find the equation of the parabola with given vertex and focus.

- (a) V(0,0) F(0,4)
- (b) V(0,0) F(0,-4)
- (c) V(0,0) F(0,3)
- (d) V(-1,2) F(5,2)

2. Find the equation of the parabola for the given focus and directrix.

- (a) F (0,6) $x=-6$
- (b) F(3,4), $x=0$
- (c) F (-2,8) $y=5$
- (d) F (5,7) $y=-9$
- (e) F(2,1) $y=5+3x$

3. Find the equation of the tangent and the normal to the parabola at the given point.

- (a) $y^2 - 4y - 4x + 8 = 0$ P (5/4,1)
- (b) $x^2 - 12y - 6 = 0$ P (4,5/6)
- (c) $y^2 - 2y + 8x + 9 = 0$ P (1,1)
- (d) $x^2 + 4y + 6y + 5 = 0$ P (3,13/3)

4. Find the equations of the tangent and normal to the parabola at the ends of the latus rectum of the given parabola.
- $y^2 - 6y + 7x - 10 = 0$
 - $x^2 + 8y + 3x - 8 = 0$
 - $y^2 - 8x + 9 = 0$
 - $x^2 + 72y + 8x + 9 = 0$
5. Find the equations of the tangent and the normal to the parabola $y^2 - 10y + 3x - 1 = 0$ at the point P (-6,3).
6. Find the parametric equation of the parabola $y^2 + 20y + 4x - 60 = 0$.
7. Find the equation of the tangent to the parabola at the given value of t.
- $y^2 - y + 3x - 10 = 0$ at $t=2$
 - $x^2 + 12y - 6x - 4 = 0$ at $t=-3$
 - $-x^2 + 4x - 2y + 22 = 0$ at $t=5$

Summary

The parabola as conic section has been studied in this chapter. The equation of the parabola has been studied as a function of focus, vertex and directrix. We have shown how to

- Get the equation given the focus and the vertex, or vertex and the directrix.
- Get the equation of the tangent and the normal to a parabola at a point on the parabola.
- Determine the parametric equation of parabola.

Further Reading

- L. Bostock, S. Chandler, and C. Rourke. **Further Pure Mathematics**. Stanley Thornes (Publishers) Ltd, 1982.
- R.E Larson and R.P. Hostetler. **Precalculus**. Houghton Mifflin Company, 1997.
- P.K. Jain and Khalil Ahmad. **Analytical Geometry of Two Dimensions**. Wiley Eastern Limited, 1983.

Chapter 5: ELLIPSE

5.0 Introduction

The ellipse was defined as the conic section whose eccentricity is less than one. This curve is also the locus of a point that moves such that the sum of the distance from two fixed points is fixed.

Objectives

At the end of the chapter, the students should be able to:-

- Determine the standard and the general equation of an ellipse given its directrix and focus or focus and eccentricity.
- Determine the parametric equation of an ellipse.
- Find the vertices and the latus rectum of an ellipse.

5.1 The equation of an ellipse

Let the focus of the ellipse be F and the directrix be DM . See figure below.

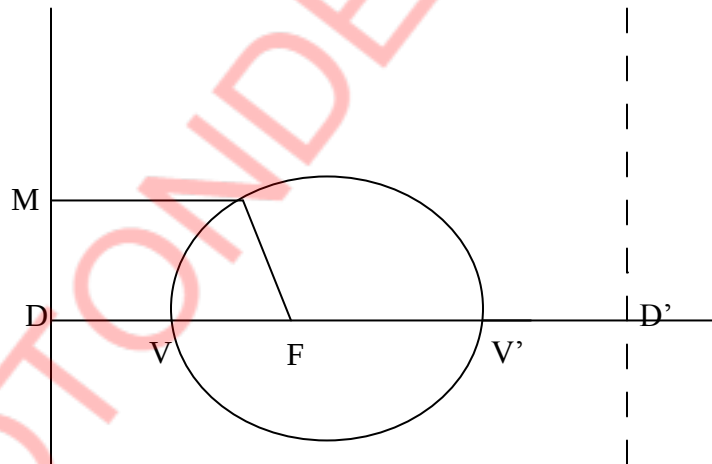


Figure 5.1

From the definition $PF = e PM$
Between D and F , there is a point V such that
 $FV = e VD$

Which lies on the ellipse.

Also further away from D along line DF there is another point V' , which satisfies
 $FV' = e V'D$

Let point C be the midpoint between V and V' . Also let the distance $CV = CV' = a$

Hence $VV' = 2a$

Then

$$FV + FV' = e(VD + V'D) = e DD' \quad \text{where } V'D' = VD \text{ also } DD' = 2CD$$

Hence

$$VV' = 2e CD = 2a \quad \text{or} \quad CD = \frac{a}{e}$$

Also write

$$FV' - FV = e(V'D - VD) = e VV' = 2ae$$

But

$$\begin{aligned} FV' - FV &= FV' + FV - 2FV \\ &= 2CV - 2FV = 2(CV - FV) \\ &= 2CF \end{aligned}$$

Hence

$$CF = ae$$

Let C be placed at the origin on the XY plane, CD be along the X-axis and the vertical line through C the Y-axis.

Notice that there now can be another point F' as the focus and directrix $D'M'$ that would have given the same results as above.

If C is the origin, the focus is $F'(ae, 0)$ and $F(-ae, 0)$ and the equation of the directrix $x = \frac{-a}{e}$

$$\text{or} \quad x' = \frac{a}{e}$$

Now

$$FP = \sqrt{(x - ae)^2 + y^2} \quad \text{and} \quad PM = \frac{a}{e} - x$$

From $FP = e PM$, we get

$$(x - ae)^2 + y^2 = \left(\frac{a}{e} - x\right)^2 e^2 \quad \text{This gives}$$

$$\begin{aligned} x^2 - 2aex + a^2 e^2 + y^2 &= e^2 \left(\frac{a^2}{e^2} - \frac{2a}{e} x + x^2 \right) \\ &= a^2 - 2aex + e^2 x^2 \end{aligned}$$

$$\Rightarrow (1 - e^2) x^2 + y^2 = a^2 - a^2 e^2 \quad \text{or} \quad \frac{x^2}{a^2} + \frac{y^2}{a^2(1 - e^2)} = 1$$

The curve has two axis. The longer one is the major axis which has length $2a$. The shorter axis obtained for the case above when $x=0$ as $2\sqrt{a^2(1 - e^2)}$ is the minor axis.

Let $b^2 = a^2(1 - e^2)$ Then the equation is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ which is in standard form.

Example 5.1

Find the directrix, eccentricity and focus of the ellipse given by $4x^2 + 9y^2 = 36$.

Solution

Writing in standard form $\frac{x^2}{9} + \frac{y^2}{4} = 1 \Rightarrow a^2 = 9 \Rightarrow a = 3$, and $b^2 = 2$.

To get the eccentricity, we have

$$b^2 = a^2(1 - e^2)$$

$$\Rightarrow 9(1 - e^2) = 4 \quad \text{or} \quad e = \sqrt{\frac{5}{9}} = \frac{\sqrt{5}}{3}$$

$$ae = 9 \times \frac{\sqrt{5}}{3} = 3\sqrt{5} \quad \text{and} \quad \frac{a}{e} = \frac{9}{\frac{\sqrt{5}}{3}} \times 3 = \frac{27}{\sqrt{5}}$$

Hence the focus is $F(3\sqrt{5}, 0)$ and the directrix is $x = \frac{27}{\sqrt{5}}$

Example 5.2

Find the equation of and sketch the ellipse whose eccentricity is $\frac{2}{3}$ center at origin and directrix $x = 6$ and focus $(1, 0)$

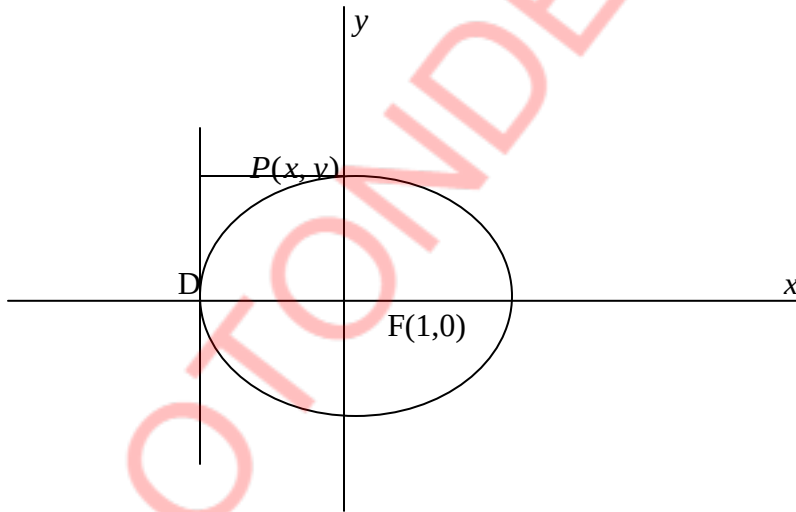
Solution

Figure 5.2

Since $\frac{a}{e} = 6$, then $a = 6e = 6 \times \frac{2}{3} = 4$.

$$b^2 = a^2(1 - e^2) = 16\left(1 - \frac{4}{9}\right) = \frac{16 \times 5}{9} = \frac{80}{9}$$

Hence the equation is $\frac{x^2}{16} + \frac{y^2}{\left(\frac{80}{9}\right)} = 1$

Exercise 5.1

- Find the eccentricity, the focus and directrix of the given ellipse.
 - $\frac{x^2}{9} + \frac{y^2}{4} = 1$
 - $\frac{x^2}{16} + \frac{y^2}{9} = 1$
 - $x^2 + 4y^2 = 9$
 - $\frac{x^2}{4} + \frac{y^2}{25} = 1$
 - $x^2 + \frac{y^2}{4} = 1$
- Find the equation of the ellipse and the directrix with center at the origin, given eccentricity and focus.
 - $e = \frac{2}{3}$ F (5,0)
 - $e = \frac{1}{5}$ F (4,0)
 - $e = \frac{1}{3}$ F (0,6)
 - $e = \frac{3}{10}$ F (0,3)
 - $e = \frac{1}{2}$ F (1,0)

5.2 The general equation of an ellipse

Using the general equation we can obtain the equation of an ellipse whose axis is not necessarily as in the illustration in section 5.2.

Consider the ellipse whose focus $F(h, k)$ and the directrix is any line $ax + by + c = 0$. Let $P(x, y)$ be any point on the ellipse.

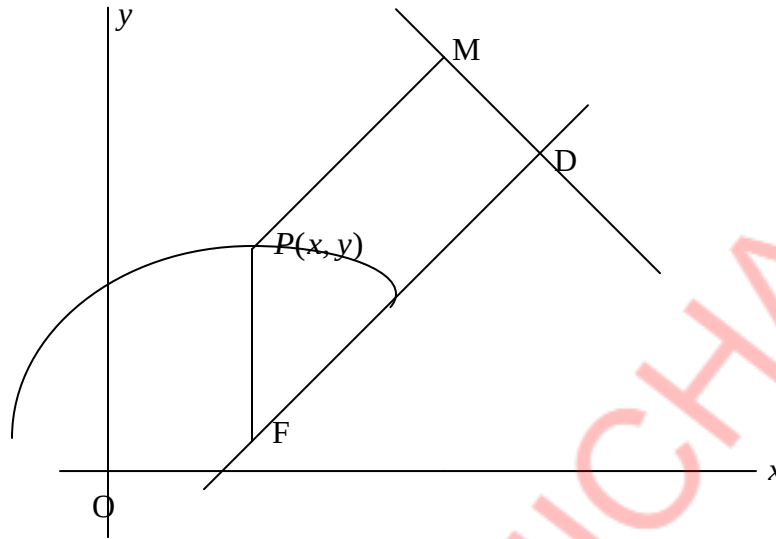


Figure 5.3

Now

$$FP = \sqrt{(x-h)^2 + (y-k)^2} \quad \text{and} \quad PM = \frac{ax+by+c}{\sqrt{a^2+b^2}}$$

Hence the general equation is

$$(x-h)^2 + (y-k)^2 = e^2 \frac{(ax+by+c)^2}{a^2+b^2}$$

Example 5.3

Find the equation of the ellipse whose focus is $\frac{2}{3}$ and the directrix is $y=9$.

Solution

From the general equation we get

$$(x-0)^2 + (y-4)^2 = \frac{4}{9}(y-9)^2 \quad \text{or} \quad x^2 + y^2 - 8y + 16 - \frac{4}{9}y^2 + 8y = 36 \quad \text{or}$$

$$x^2 + \frac{5}{9}y^2 = 20 \quad \text{which is} \quad \frac{x^2}{20} + \frac{y^2}{36} = 1$$

This is an ellipse whose major axis is along the y-axis and is of length $2b=2\sqrt{36}=12$ and minor axis along the y-axis and length $2a=2\sqrt{20}=4\sqrt{5}$.

We have seen that when the center of the ellipse is $(0,0)$, then the equation is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

If the center is shifted to say $C(m,n)$, the equation of the ellipse becomes

$$\frac{(x-m)}{a^2} + \frac{(y-n)}{b^2} = 1$$

Example 5.4

Find the focus, the center, the eccentricity and directrix of the ellipse whose equation is $x^2 + 4y^2 + 4x - 24y + 24 = 0$

Solution

Writing in the standard form gives

$$x^2 + 4y^2 + 4x - 24y = -24 \quad \text{or} \quad (x+2)^2 + 4(y-3)^2 = -24 + 4 + 36 = 16$$

Hence

$$\frac{(x+2)^2}{16} + \frac{(y-3)^2}{4} = 1 \quad \text{Where} \quad a^2 = 16 \Rightarrow a=4 \quad b^2 = 4 \Rightarrow b=2$$

The center is C (-2,3) eccentricity is given by $b^2 = a^2(1-e^2) \Rightarrow 4 = 16(1-e^2)$ or $e = \frac{\sqrt{3}}{2}$

Since the centre is C (-2,3) the abscissa for the foci are $-2 \pm ae = -2 \pm \frac{\sqrt{3} \cdot 4}{2}$

Here they are

$$F_1(-2-2\sqrt{3}, 3) \quad \text{and} \quad F_2(-2+2\sqrt{3}, 3)$$

The directrix is at

$$x = -2 \pm \frac{a}{e} = -2 \pm \frac{8}{\sqrt{3}}$$

5.3 The length of the latus rectum.

Consider the origin-centered ellipse.

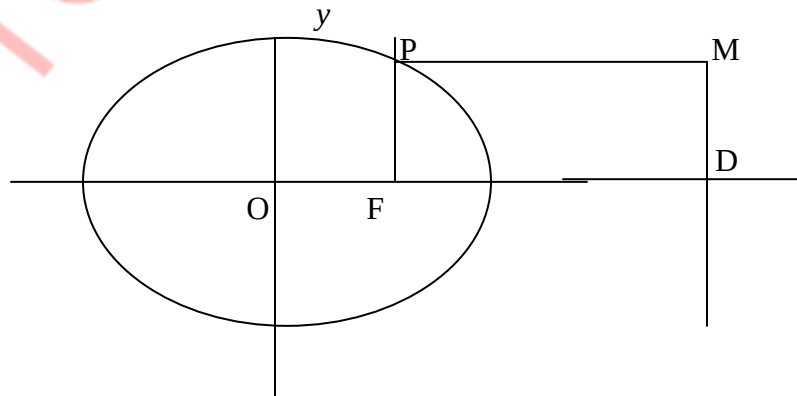


Figure 5.4

From the definition of an ellipse we have for the length l through the focus

$$\frac{l}{\frac{a}{e} - ae} = e \Rightarrow l = e\left(\frac{a}{e} - ae\right) \quad \text{Or} \quad l = a(1-e)^2$$

Hence the length of an ellipse is $2l = 2a(1-e)^2$

Exercise 5.2

1. Find the equation of the ellipse with the given focus, directrix and eccentricity.

(a) F (2,1) directrix $x = 2$, $e = \frac{1}{2}$

(b) F (4,7) directrix $y = 2$, $e = \frac{1}{4}$

(c) F (-2,3) directrix $y = 7$, $e = \frac{3}{5}$

(d) F (-1,1) directrix $x + y - 2 = 0$, $e = \frac{2}{7}$

(e) F (4,10) directrix $3x + 8y - 2 = 0$, $e = \frac{1}{2}$

2. Find the foci, the center, the eccentricity and directrix of the given ellipse.

(a) $\frac{x^2}{169} + \frac{y^2}{25} = 1$

(b) $9x^2 + 4y^2 + 36x - 24y + 36 = 0$

(c) $16x^2 + 25y^2 - 32x + 50y + 16 = 0$

(d) $4x^2 + 9y^2 - 48x + 72y + 144 = 0$

(e) $5x^2 + 9y^2 - 80x + 5y + 221 = 0$

3. Find the equation of the ellipse with its center at (1,2), focus at (6,2) and passing through point (4,6).

4. Find the vertices and the length of the latus rectum of the ellipse,

(a) $\frac{x^2}{8} + \frac{y^2}{4} = 1$

(b) $\frac{x^2}{16} + \frac{y^2}{12} = 1$

(c) $225x^2 + 289y^2 = 65025$

(d) $36x^2 + 11y^2 - 144x - 144y - 208 = 0$

(e) $3x^2 + 4y^2 - 28x - 16y + 48 = 0$

5.4 The tangent of an ellipse.

Consider the ellipse whose equation is $\frac{(x-m)^2}{a^2} + \frac{(y-n)^2}{b^2} = 1$

We wish to get the equation of the ellipse at point say $P_1(x_1, y_1)$ on the ellipse.

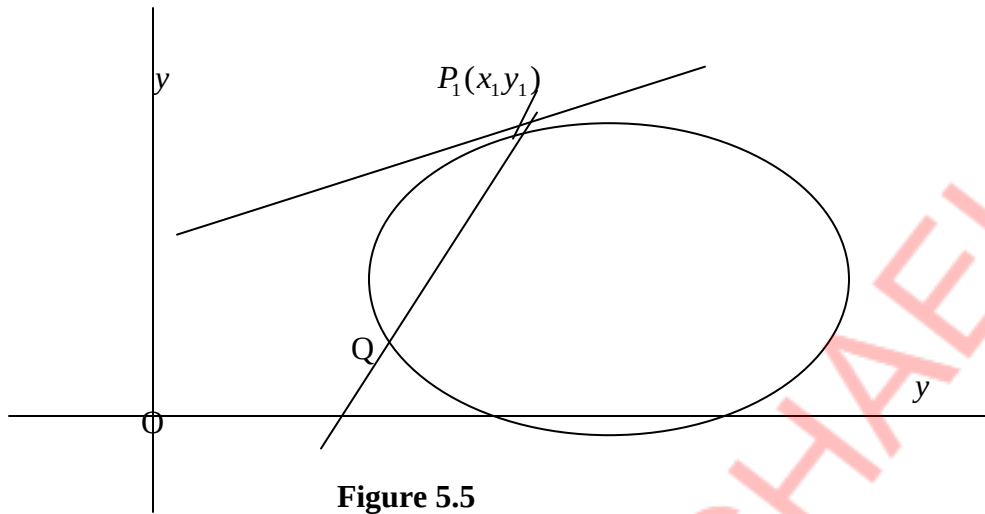


Figure 5.5

Let $Q(x_2, y_2)$ be any other point on the ellipse which approaches P_1 . The gradient of P_1Q_1 is $m \cdot \frac{y_2 - y_1}{x_2 - x_1}$. As Q approaches P_1 we get the limit from: since Q and P_1 are on the ellipse, then we have

$$\frac{(x_1 - m)^2}{a^2} + \frac{(y_1 - n)^2}{b^2} = 1 \quad \text{--- 1 ---}$$

And

$$\frac{(x_2 - m)^2}{a^2} + \frac{(y_2 - n)^2}{b^2} = 1 \quad \text{--- 2 ---}$$

Subtracting 1 from 2 we get

$$\frac{(x_2 - m)^2 - (x_1 - m)^2}{a^2} + \frac{(y_2 - n)^2 - (y_1 - n)^2}{b^2} = 0$$

Or

$$(x_2 - x_1)(x_2 + x_1 - 2m) = \frac{-a^2}{b^2} (y_2 - y_1)(y_1 - n + y_2 - n)$$

$$\Rightarrow \frac{y_2 - y_1}{x_2 - x_1} = \frac{-b^2}{a^2} \frac{(x_1 - m + x_2 - m)}{(y_1 - n + y_2 - n)}$$

With the limit

$$\lim_{Q \rightarrow P} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) = \frac{-b^2}{a^2} \left(\frac{x_1 - m}{y_1 - n} \right)$$

Hence the equation of the tangent is for any point $P(x, y)$ on the tangent

$$\frac{y - y_1}{x - x_1} = \frac{-b^2}{a^2} \frac{(x_1 - m)}{(y_1 - n)}$$

$$\text{Or} \quad (y - y_1)(y_1 - n) = \frac{-b^2}{a^2} (x - x_1)(x_1 - m)$$

This can be written as

$$\begin{aligned} [y - n - (y_1 - n)](y_1 - n) &= \frac{-b^2}{a^2} [x - m - (x_1 - m)](x_1 - m) \\ \Rightarrow b^2(x - m)(x_1 - m) + a^2(y - n)(y_1 - n) &= b^2(x_1 - m)^2 + a^2(y_1 - n)^2 = a^2b^2 \end{aligned}$$

Or

$$\frac{(x - m)(x_1 - m)}{a^2} + \frac{(y - n)(y_1 - n)}{b^2} = 1$$

This is the equation of the tangent.

Example 5.5

Find the equation of the tangent and equation of the normal at point $P(-1, 2)$ of the ellipse $(x + 1)^2 + 4(y - 1)^2 = 4$.

Solution

Write the equation in the standard form to get

$$\frac{(x + 1)^2}{4} + \frac{(y - 1)^2}{1} = 1$$

The tangent passes through $(-1, 2) \Rightarrow x_1 = -1$ and $y_1 = 2$. We get the equation of the tangent as

$$\frac{(x + 1)(-1 + 1)}{4} + \frac{(y - 1)(2 - 1)}{1} = 1 \Rightarrow y = 5$$

5.5 Parametric equation of an ellipse

To get the parametric equation of an ellipse we first define the auxiliary circle. The auxiliary circle of an ellipse is the circle whose center is the center of the ellipse and its diameter is equal to the length of the major axis.

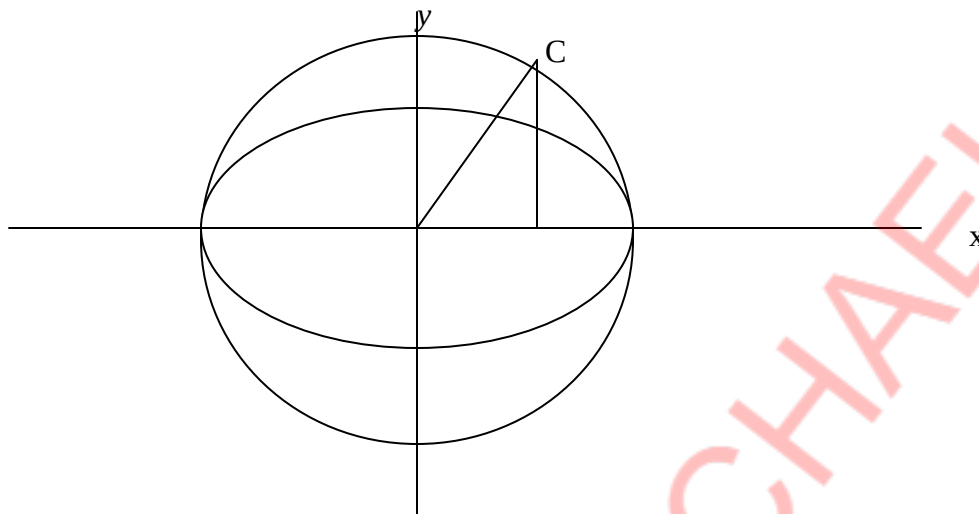


Figure 5.6

Let ϕ be any angle on the auxiliary circle as in the figure above. Consider point $P(x, y)$ on the ellipse and on the vertical line through C. Angle ϕ is called the eccentric angle of point $P(x, y)$.

Obviously $x = a \cos \phi$. To get y substitute in the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

to get

$$\frac{a^2 \cos^2 \phi}{a^2} + \frac{y^2}{b^2} = 1$$

which comes to $y = b \sin \phi$. Hence the parametric equations of an ellipse are

$$x = a \cos \phi, y = b \sin \phi.$$

$(a \cos \phi, b \sin \phi)$ is a point on the ellipse.

Example 5.6

Find the equation of the chord joining two points whose eccentric angles are ϕ_1 and ϕ_2 on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Solution

Let the two points be $P_1(a \cos \phi_1, b \sin \phi_1)$ and $P_2(a \cos \phi_2, b \sin \phi_2)$. The equation of chord P_1P_2 is

$$y - b \sin \phi_1 = \frac{b \sin \phi_2 - b \sin \phi_1}{a \cos \phi_2 - a \cos \phi_1} (x - a \cos \phi_1)$$

$$\Rightarrow y - b \sin \phi_1 = \frac{b \cos[(\phi_2 + \phi_1)/2]}{-a \sin[(\phi_2 + \phi_1)/2]} (x - a \cos \phi_1)$$

This reduces to

$$bx \cos\left(\frac{\phi_1 + \phi_2}{2}\right) + ay \sin\left(\frac{\phi_1 + \phi_2}{2}\right) = ab \cos\left(\frac{\phi_1 - \phi_2}{2}\right).$$

Hence the equation of the chord joining the two points is

$$\frac{x}{a} \cos\left(\frac{\phi_1 + \phi_2}{2}\right) + \frac{y}{b} \sin\left(\frac{\phi_1 + \phi_2}{2}\right) = \cos\left(\frac{\phi_1 - \phi_2}{2}\right)$$

Exercise 5.3

- Find the equation of the tangent and the normal at the given point of the given ellipse.

(a) $\frac{x^2}{25} + \frac{y^2}{9} = 1$, $\left(2, \frac{1}{5}\sqrt{189}\right)$

(b) $4x^2 + 9y^2 = 144$, $(-3, 2\sqrt{3})$

(c) $x^2 + 5y^2 - 2x + 20y - 16 = 0$, $\left(1 + \frac{1}{2}\sqrt{23}, \frac{1}{2}\right)$

(d) $13x^2 + 23y^2 - 51x - 19y - 4 = 0$, $(1, -1)$

- Find the parametric equation of the given ellipse.

(a) $\frac{x^2}{36} + \frac{y^2}{4} = 1$

(b) $x^2 + 2y^2 - 2x + 8y + 5 = 0$

(c) $x^2 + 9y^2 + 10x - 54y + 97 = 0$

(d) $\frac{x^2}{4} + \frac{y^2}{25} = 1$

Miscellaneous Exercises

- Find the equation of the ellipse whose center is origin, eccentricity and focus given.

(a) $e = \frac{1}{6}$, F (9,0)

(b) $e = \frac{9}{10}$, F (0,7)

(c) $e = \frac{2}{7}$, F (7,0)

(d) $e = \frac{1}{12}$, F (0,12)

- Find the foci, the vertices, eccentricity, length of the latus rectum and equation of the directrix for the given ellipse.

- (a) $\frac{x^2}{16} + \frac{y^2}{4} = 1$
 (b) $9x^2 + 36y^2 = 324$
 (c) $4x^2 + 8x + 9y^2 - 72y + 112 = 0$
 (d) $9x^2 + 90x + 4y^2 - 16y + 205 = 0$
3. Find the tangent to the ellipse $x^2 + 4y^2 = 4$ at point $\left(1, -\frac{1}{2}\sqrt{3}\right)$
4. Find the length of the latus rectum of the ellipse $5x^2 + 60x + 3y^2 + 12y + 191 = 0$
5. Determine the equation of the ellipse which is the locus of the point that moves such that the sum of its distances from the points (4,5) and (9,5) is 15 units. Find the vertices, eccentricity and length of the latus rectum.
6. The eccentric circle of an ellipse is $x^2 + y^2 - 4x - 6y - 12 = 0$. If its eccentricity is $e = \frac{2}{3}$, and the major axis horizontal, find the foci, the vertices and length of latus rectum.
7. Find the parametric equation of the ellipse $16x^2 + 9y^2 + 32x - 36y - 92 = 0$
8. Find the vertices, foci, length of the latus rectum and the equation of directrix of the parabola $x = 7 \cos \phi, y = 3 \sin \phi$

Summary

In this chapter, we have studied the hyperbola. We have shown how to:

- Determine the general and the standard equation of an ellipse
- Find the tangent to an ellipse at a given point on the ellipse
- Find the parametric equation of an ellipse.

Further Reading

- L. Bostock, S. Chandler, and C. Rourke. **Further Pure Mathematics**. Stanley Thornes (Publishers) Ltd, 1982.
- R.E Larson and R.P. Hostetler. **Precalculus**. Houghton Mifflin Company, 1997.
- P.K. Jain and Khalil Ahmad. **Analytical Geometry of Two Dimensions**. Wiley Eastern Limited, 1983.

Chapter 6: HYPERBOLA

6.0 Introduction

The hyperbola was briefly described in section - - - as the locus of a point that moves such that the ratio of its distance from a fixed point to the distance from a fixed line is greater than one.

Objectives

At the end of the chapter, the students should be able to:-

- (a) Determine the standard and the general equation of a hyperbola.
- (b) Find the tangents and the normal to the hyperbola at a given point on the hyperbola.
- (c) Determine the asymptotes of a hyperbola.
- (d) Find the parametric equation of a hyperbola.

6.1 Equation of a hyperbola

The standard equation can be developed as follows:

Consider a fixed point F and a fixed line D as in the diagram below.

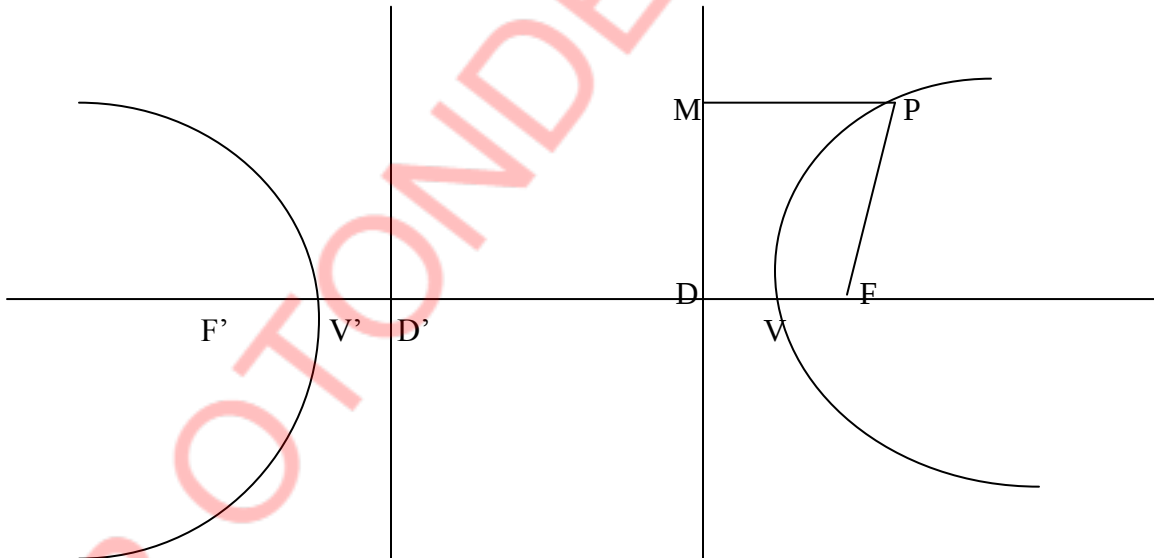


Figure 6.1

$P(x, y)$ is on the parabola such that $\frac{PF}{MP} = e > 1$ or $PF = eMP$

Let the axis of the hyperbola be the horizontal line perpendicular to the directrix through the focus and vertex V be the point of intersection of the hyperbola and the axis,

Hence

$$FV = eVD.$$

There also exists another point V_1 along the axis such that $FV' = eV'D$. Infact there is a whole set of points and also the virtual focus F and directrix D .

Let the distance $VV' = 2a$ and let C be the midpoint of VV' .

Then

$$FV' - FV = e(V'D - VD) = e(D'D = e2CD$$

But

$$FV' - FV = V'V = 2a$$

$$\Rightarrow 2eCD = 2a \Rightarrow CD = \frac{a}{e}$$

Also

$$FV' + FV = e(V'D + VD)eV'V = 2ae$$

But

$$FV' + FV = F'F = 2CF \Rightarrow 2CF = 2ae \Rightarrow CF = ae$$

Let C be at the origin. Then the foci are $F(ae, 0)$ and $F'(-ae, 0)$. The directrix is $x = \frac{a}{e}$.

From $PF = ePM$, $PF = \sqrt{(x - ae)^2 + y^2}$, $PM = x - \frac{a}{e}$, we get

$$(x - ae)^2 + y^2 = e^2 \left(x - \frac{a}{e}\right)^2,$$

or

$$x^2(e^2 - 1) - y^2 = a^2(e^2 - 1),$$

or

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

where $b^2 = a^2(e^2 - 1)$. This is the standard equation of a hyperbola.

The curve has two lines of symmetry, the x-axis and the y-axis. VV' is called the transverse axis and BB' the conjugate axis.

The difference between PV' and PV is a constant equal to $2a$ as shown below.

$$PF = e\left(x - \frac{a}{e}\right) = ex - a$$

$$PF' = e\left(x + \frac{a}{e}\right) = ex + a$$

$$\Rightarrow PF' - PF = 2a.$$

This provides an alternative definition of a hyperbola as the locus of a point that moves such that the difference between the distances from two fixed points a constant.

Example 6.1

Determine the equation of the hyperbola whose eccentricity is $3/2$ and the foci are $F'(-2,0)$ and $F(2,0)$.

Solution

Since $ae = 2$, and $e = 3/2$, then $a = 4/3$. This implies

$$b^2 = a^2(e^2 - 1) = \frac{16}{9}\left(\frac{9}{4} - 1\right) = \frac{20}{9}.$$

Hence the equation is

$$\frac{x^2}{(16/9)} - \frac{y^2}{(20/9)} = 1.$$

The latus rectum is the chord perpendicular to the transverse axis and through the focus. The length of the latus rectum is obtained as follows. Let the length be $l = 2l_1$, where l_1 is the distance from the focus along the latus rectum to the curve. Then

$$\frac{l_1}{ae - a/e} = e, \Rightarrow l_1 = a(e^2 - 1).$$

Hence the length of the latus rectum is

$$l = 2l_1 = 2a(e^2 - 1).$$

Example 6.2

Find the coordinates of the foci, the eccentricity, the length of the latus rectum and the directrix of the hyperbola

$$\frac{x^2}{36} - \frac{y^2}{4} = 1.$$

Solution

From the equation $a = 6$ and $b = 2$. The eccentricity is obtained from $b^2 = a^2(e^2 - 1)$ as $e = \frac{1}{3}\sqrt{10}$.

The foci are therefore $F'(-3\sqrt{10}, 0)$ and $F(3\sqrt{10}, 0)$. The length of the latus rectum is

$$L = 2a(e^2 - 1) = 2 \times 9\left(\frac{10}{9} - 1\right) = 2$$

Exercise 6.1

1. Find the equation of the hyperbola with center at the origin, the given focus and eccentricity.

(a) Focus $F(6, 0)$, eccentricity $e = \frac{5}{2}$

- (b) Focus $F(1,0)$, eccentricity $e = \frac{5}{2}$
- (c) Focus $F(5,0)$, eccentricity $e = \frac{7}{4}$
- (d) Focus $F(0,3)$, eccentricity $e = \frac{6}{5}$
- (e) Focus $F(0,2)$, eccentricity $e = \frac{7}{2}$
- (f) Focus $F(0,8)$, eccentricity $e = \frac{10}{7}$
2. Find the foci, the vertices, the eccentricity, and the equation of the directrix of the given hyperbola.
- (a) $\frac{x^2}{9} - \frac{y^2}{4} = 1$
- (b) $\frac{x^2}{25} - \frac{y^2}{36} = 1$
- (c) $x^2 - 9y^2 = 9$
- (d) $9x^2 - 16y^2 = 144$
- (e) $49x^2 - 16y^2 = 784$
- (f) $\frac{y^2}{9} - \frac{x^2}{4} = 1$
- (g) $25y^2 - 4x^2 = 100$

6.2 The Asymptotes of a hyperbola

Consider the hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and let $y = mx + C$ be any line. The line and hyperbola meet at points where

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{x^2}{a^2} - \frac{(mx + C)^2}{b^2} = 0$$

$$\Rightarrow (b^2 - m^2 a^2)x^2 - 2a^2 mCx - a^2(b^2 + c^2) = 0$$

The points of intersection are at infinity if the coefficients of x^2 and x are zero.

This implies $b^2 - m^2 a^2 = 0$

$$\Rightarrow m = \pm \frac{b}{a} \quad \text{and} \quad 2a^2 mC = 0 \Rightarrow c = 0.$$

Hence the asymptotes of the hyperbola are $y = \pm \frac{b}{a} x$

Exercise 6.2

1. Find the length of the latus rectum and the asymptotes of the hyperbola given.

(a) $9x^2 - 4y^2 = 36$

(b) $\frac{x^2}{5} - \frac{y^2}{3} = 1$

(c) $7y^2 - 4x^2 = 28$

(d) $4y^2 - x^2 = 16$

(e) $x^2 - 4y^2 = 4$

6.3 Hyperbola with center not at origin

Suppose the center of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is shifted to position $C(h, k)$. The equation becomes

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

If the transverse axis is along the y-axis, then the equation becomes

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

Example 6.4

Analyze the hyperbola $9x^2 - 16y^2 - 18x - 64y - 199 = 0$.

Solution

Complete the squares and write the equation in standard form to get

$$9(x-1)^2 - 16(y+2)^2 = 144$$

or

$$\frac{(x-1)^2}{16} - \frac{(y+2)^2}{9} = 1.$$

This is the equation of the hyperbola with center $C(1, -2)$. Also $a=4$ and $b=3$. The eccentricity is

$$e = \sqrt{\frac{b^2}{a^2} + 1} = \sqrt{\frac{9}{16} + 1} = \frac{5}{4} \Rightarrow ae = 5.$$

The vertices are $V'(-3, -2)$ and $V(5, -2)$, foci $F'(-4, -2)$ and $F(6, -2)$, asymptotes

$$y + 2 = \pm \frac{3}{4}(x - 1).$$

6.4 Tangent and normal to a hyperbola

Consider the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

The equation of the tangent to the hyperbola at point $P(x_1, y_1)$ is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

Note that the normal at any point on the curve can be easily obtained since it is the line perpendicular to the tangent at that point.

Example 6.3

Determine the eccentricity, the foci, the length of the latus rectum, the equation of the asymptotes, the equation of the tangent and the normal at point $P(3, \frac{12}{5})$ of the hyperbola

$$\frac{x^2}{25} - \frac{y^2}{9} = -1$$

Solution

From the equation, $a = 5$, $b = 3$ Hence from $b^2 = a^2(e^2 - 1)$ we get the eccentricity as

$$e = \sqrt{1 + \frac{9}{25}} = \sqrt{\frac{34}{25}}, \text{ and } ae = \sqrt{34}.$$

Hence the foci are $F'(-\sqrt{34}, 0)$ and $F(\sqrt{34}, 0)$.

The length of the latus rectum is

$$L = 2a(e^2 - 1) = 2 \times 5 \left(\frac{34}{25} - 1 \right) = \frac{18}{5}.$$

The asymptotes are $y = \pm \frac{b}{a}x$. Hence $y = \frac{3}{5}x$ and $y = -\frac{3}{5}x$.

The equation of the tangent at $P(3, \frac{12}{5})$ is

$$\frac{3}{25}x - \frac{12}{45}y = 1, \Rightarrow y = \frac{3}{10}x - \frac{15}{4}.$$

The normal has therefore slope $m = \frac{-10}{3}$. Since it passes through P , then the equation is

$$\frac{y - 12/5}{x - 3} = \frac{-10}{3}, \Rightarrow y = \frac{57}{25} - \frac{10}{3}x.$$

6.6 Parametric equation of a hyperbola

To get the parametric equation of a hyperbola, consider the auxiliary circle whose center is the center of the hyperbola and the diameter is the length of the transverse axis, $2a$. Let $P(x, y)$ be a point on the hyperbola.

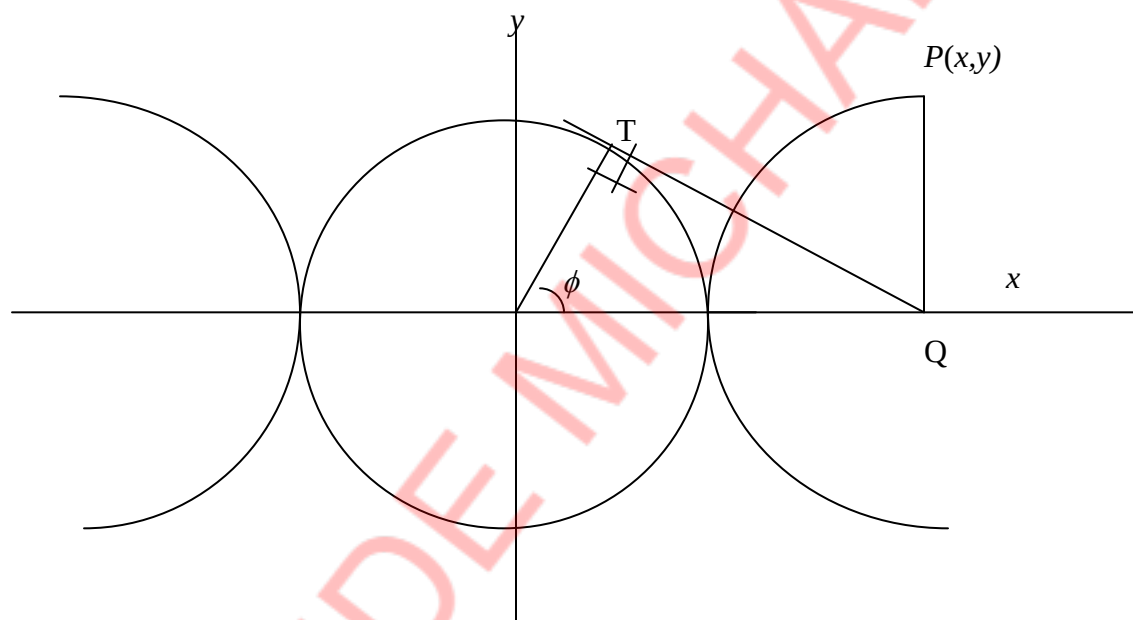


Figure 6.2

Let TQ be the tangent to the circle at point T and Q be a point on the x-axis such that $\overline{OQ} = x$. Then

$$x = \frac{a}{\cos \phi}$$

To get y, put

$$\left(\frac{a}{\cos \phi} \right)^2 \frac{1}{a^2} - \frac{y^2}{b^2} = 1$$

which reduces to

$$y = b \tan \phi.$$

The parametric equations of a hyperbola is therefore

$$x = a \sec \phi, y = b \tan \phi$$

The point $(a \sec \phi, b \tan \phi)$ is a point on the parabola. Note that when $-\pi/2 < \phi < \pi/2$ the parametric equation describes the right half of the curve. When $\pi/2 < \phi < 3\pi/2$ the left half is described. When $\phi = \pi/2$ and $\phi = -\pi/2$, the point would be at the asymptote.

Example 6.5

Determine the point on the directrix of the hyperbola $x = 7 \sec \phi, y = 2 \tan \phi$ which the tangent at the point $(7 \sec 60, 2 \tan 60)$ passes through.

Solution

From the parametric equation, $a=7$ and $b=2$. Hence the Cartesian equation of the hyperbola is

$$\frac{x^2}{49} - \frac{y^2}{4} = 1$$

The eccentricity is

$$e = \sqrt{\frac{b^2}{a^2} + 1} = \sqrt{\frac{4}{49} + 1} = \frac{\sqrt{53}}{7}$$

The equation of the directrix is therefore

$$x = \frac{a}{e} = \frac{49}{\sqrt{53}}$$

Point $(7 \sec 60, 2 \tan 60)$ is point $(14, 2\sqrt{3})$. The tangent to hyperbola at this point has equation

$$\frac{14}{49}x - \frac{2\sqrt{3}}{4}y = 1, \Rightarrow y = \left(\frac{2}{7}x - 1\right)\frac{\sqrt{3}}{2}$$

Substituting $x = \frac{49}{\sqrt{53}}$ gives

$$y = \left(\frac{14}{\sqrt{53}} - 1\right)\frac{\sqrt{3}}{2} \approx 0.8.$$

Hence the tangent at $(7 \sec 60, 2 \tan 60)$ intersects with the directrix at $(1.9, 0.8)$.

Exercise 6.5

1. Find the vertices, the foci, the eccentricity, the directrix, the length of the latus rectum and the asymptotes of the given hyperbola.

(a) $x^2 - 4y^2 - 6x - 16y - 11 = 0$

(b) $25y^2 - 9x^2 - 225 = 0$

(c) $x^2 - 9y^2 + 36y - 72 = 0$

(d) $3y^2 - 5x^2 = 15$

(e) $9x^2 - y^2 + 54x + 10y + 55 = 0$

2. Find the equation of the tangent at the given point on the hyperbola.

(a) $4x^2 - 9y^2 = 36, \quad \left(2, \frac{2}{3}\sqrt{5}\right)$

(b) $25y^2 - x^2 = 25, \quad (5, \sqrt{2})$

- (c) $x^2 - 4y^2 + 6x + 16y - 11 = 0$, $\left(10, 2 + \frac{1}{8}\sqrt{423}\right)$
- (d) $2x^2 - 3y^2 = 6$, $\left(2, \sqrt{\frac{2}{3}}\right)$
- (e) $4x^2 - 9y^2 = 576$, $(15, 6)$
3. Find the parametric equation of the hyperbola with vertices and foci as given.
- (a) $V_1(-2, 0)$ $V_2(2, 0)$ $F_1(-3, 0)$ $F_2(3, 0)$
- (b) $V_1(-6, 0)$ $V_2(-6, 0)$ $F_1(-12, 0)$ $F_2(12, 0)$
- (c) $V_1(0, -2)$ $V_2(0, 2)$ $F_1(0, -4)$ $F_2(0, 4)$
- (d) $V_1(0, -1)$ $V_2(0, 1)$ $F_1(0, -5)$ $F_2(0, -5)$
4. Determine the equation of the tangent at the given point on the hyperbola and also obtain the focal distance of the point.
- (a) $x = 8\sec\phi$, $y = \tan\phi$, $(8\sec 30, 3\tan 30)$
- (b) $x = 4\sec\phi$, $y = 2\tan\phi$, $(4\sec 45, 2\tan 45)$
- (c) $x = 10\sec\phi$, $y = 7\tan\phi$, $(10\sec 10, 7\tan 10)$
- (d) $x = 20\sec\phi$, $y = \tan\phi$, $(20\sec 60, \tan 60)$

Miscellaneous Exercises

1. Find the vertices, length of the transverse axis, foci and eccentricity of the hyperbola.
- (a) $\frac{x^2}{16} - y^2 = 1$
- (b) $\frac{x^2}{49} - \frac{y^2}{16} = 1$
- (c) $\frac{y^2}{12} - \frac{x^2}{5} = 1$
- (d) $9x^2 - 4y^2 = 36$
2. Find the equation of the hyperbola with horizontal axis given center, eccentricity and the directrix.
- (a) Center $(3, 5)$; $e = 3/2$; directrix $x = 5$
- (b) Center $(-2, -8)$; $e = 5/2$; directrix $x = 4$
- (c) Center $(-5, 2)$; $e = 7/3$; directrix $x = 0$
- (d) Center $(6, 0)$; $e = 4/3$; directrix $x = 8$.
3. Find the length of latus rectum and the asymptotes of the given parabola.
- (a) $3x^2 - 18x - 4y^2 + 8y + 22 = 0$
- (b) $x^2 - 2y^2 + 4x - 12y + 20 = 0$
- (c) $25x^2 - 16y^2 = 400$
- (d) $21x^2 - 3y^2 - 126x + 186 = 0$
4. Find the equation of the tangent and the normal to the hyperbola
- $$25x^2 - 9y^2 - 200x - 36y + 139 = 0$$

- at the point $(4 + \frac{1}{5}\sqrt{216}, -3)$.
5. Find the equation of the hyperbola with the center at the origin, axis on the coordinate axis, and passes through points (3,1) and (9,5).
 6. Find the parametric equation of the hyperbola $6x^2 - 5y^2 + 72x - 10y + 210 = 0$.

Summary

In this chapter, we have studied the hyperbola. We have shown how to:

- (a) Determine the general and the standard equation of a hyperbola.
- (b) Find the asymptotes of a hyperbola.
- (c) Find the tangent to a hyperbola at a given point on the hyperbola.
- (d) Find the parametric equation of a hyperbola.

Further Reading

- 1 L. Bostock, S. Chandler, and C. Rourke. **Further Pure Mathematics**. Stanley Thornes (Publishers) Ltd, 1982.
- 2 R.E Larson and R.P. Hostetler. **Precalculus**. Houghton Mifflin Company, 1997.
- 3 P.K. Jain and Khalil Ahmad. **Analytical Geometry of Two Dimensions**. Wiley Eastern Limited, 1983.

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Chapter 7: POLAR COORDINATES

7.0 Introduction

In the previous chapters the study of analytical geometry was based on the rectangular coordinate system. In this chapter we introduce the polar coordinate system. In the system, a fixed point is described by its distance from a fixed point and its direction from a fixed line through the point. The fixed point is called the pole and the fixed line is called the polar axis.

Objectives

By the end of the chapter, the students should be able to:

- (a) Describe the polar coordinates of a point in relation to the pole and the polar axis.
- (b) Determine the polar equation of a line and non linear curves.
- (c) Convert a Cartesian equation to polar and vice versa.

7.1 The polar coordinates of a point

Let the fixed point be O (Fig). Suppose point P is at a distance r from O . Let angle XOP be θ . Then the polar coordinates of P are given as an ordered pair (r, θ)

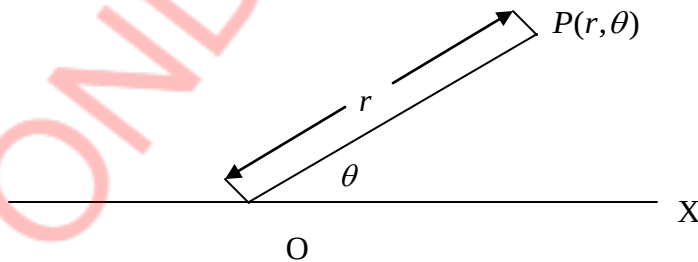


Figure 7.1

The angle is measured in an anticlockwise direction from the fixed line OX . A negative value of θ implies the angle is measured clockwise from the fixed line. The distance r is always positive as long as it is measured from the O to P .

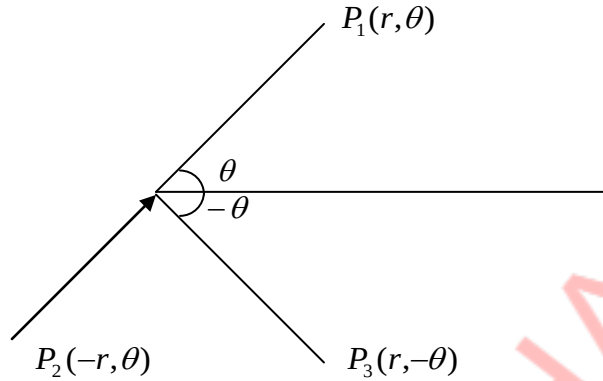


Figure 7.2

Points $P_1(r, \theta)$, $P_2(-r, \theta)$ and $P_3(r, -\theta)$ give the various positions when r and θ take negative values.

7.2 Relationship between polar and rectangular coordinates

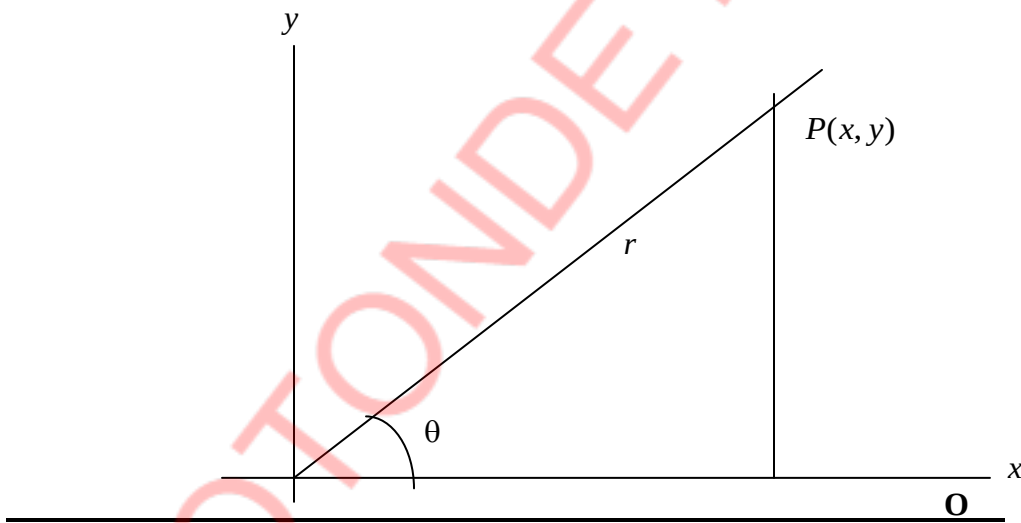


Figure 7.3

Consider a point $P(x, y)$ whose polar coordinates are $P(r, \theta)$. From the diagram it is clear that

$$x = r \cos \theta, \text{ and } y = r \sin \theta \qquad \frac{y}{x} = \tan \theta.$$

$$r^2 = x^2 + y^2$$

The above relationships can be used to transform the Cartesian coordinates to polar coordinates and vice versa.

Example 7.1

Consider point (3,4) on the x y plane. From the above relationships, we get

$$r = \sqrt{3^2 + 4^2} = 5 \text{ and } \theta = \tan^{-1} \frac{4}{3} = 53.13$$

Hence the polar coordinates for $P(3,4)$ are $P(5,53.13)$

Example 7.2

Consider point (7,60) on the plane. From the above relationships, we get $r = 5$,

$$x = r \cos \theta = 7 \cos 60 = 7/2$$

$$y = r \sin \theta = 7 \sin 60 = 7\sqrt{3}/2$$

Hence the polar coordinates for (7,60) are $(7/2, 7\sqrt{3}/2)$.

7.3 Polar equation of a line

The angle of a line which will pass through the origin is fixed at say α where α is independent of r . The part that lies on the positive side of the initial line from O is called the part-line with equation $\theta = \alpha$

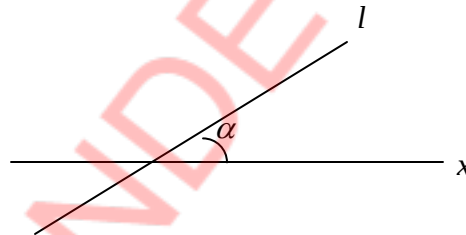


Figure 7.4

Suppose the line l does not pass through the origin but is at distance p from the origin

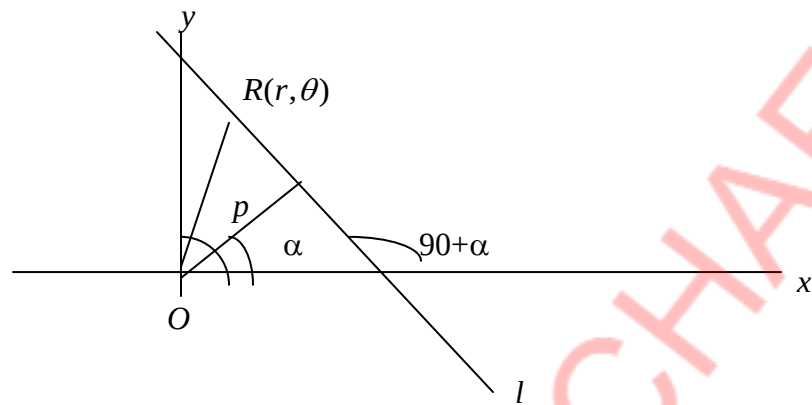


Figure 7.5

Let $R(r, \theta)$ be any point on the line and α be the angle made by the perpendicular from origin to the line. Then we see that if $ON = p$, then

$p = r \cos(\theta - \alpha)$, the angle made by the line to the normal line is $90 + \alpha$

Here the equation of a line in polar form is

$$p = r \cos(\theta - \alpha)$$

7.4 Polar equation of a circle

If the radius of the circle is $r = a$, and the center at the origin, then the equation of the circle is simply given as $r = a$. θ varies from zero to 360 degrees.

Suppose the center of the circle of the circle is $C(r_1, \theta_1)$ and let $P(r, \theta)$ be any point on the circle. Let the radius of the circle be a . Then

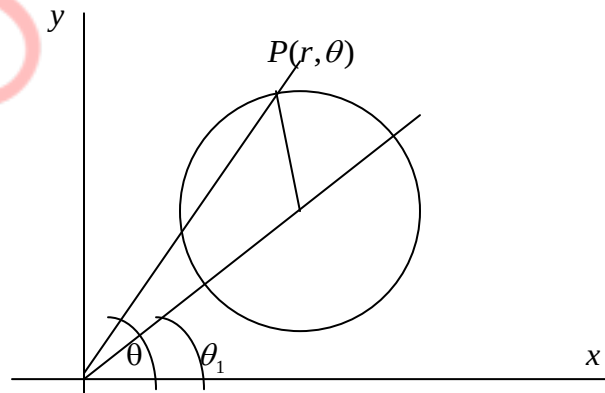


Figure 7.6

Then from triangle OCP we have

$$OC^2 + OP^2 - 2 \overline{OC} \times OP \cos \angle COP = a^2 = \overline{CP}^2$$

$$\angle COP = \theta - \theta_1$$

Here $a^2 = r_1^2 + r^2 - 2rr_1 \cos(\theta - \theta_1)$

This is the polar equation of a circle.

Example 7.3

Suppose the radius of a circle is a . Let the center be $(b,0)$. Then the equation of the circle is $b^2 + r^2 - 2br \cos \theta = a^2$.

If the circle passes through the pole with center $(b,0)$, then the equation of the circle is $r = 2a \cos \theta$.

Example 7.4

Find the polar equation of the circle whose center has the Cartesian coordinates $C(5,8)$ and radius is 10.

Solution

First find θ_1 which is given by

$$\theta_1 = \tan^{-1}(8/5) = 58.$$

Also

$$r_1 = \sqrt{8^2 + 5^2} = \sqrt{89}$$

Hence

$$10^2 = 89 + r^2 - 2r\sqrt{89} \cos(\theta - 58).$$

Or

$$r^2 - 2r\sqrt{89} \cos(\theta - 58) = 11$$

7.5 Polar equation of a conic.

To get the polar equation of conic, recall a conic was defined as the locus of a point that moves such that its distance from fixed point, the focus, is a fixed ratio to its distance to a fixed line, the directrix.

For illustration, we consider the case when the focus is at the origin. The polar equation takes a slightly different form depending on which side of the focus the directrix is as will be shown.

Consider the case when the directrix is at the left of the focus and the axis being the x -axis.

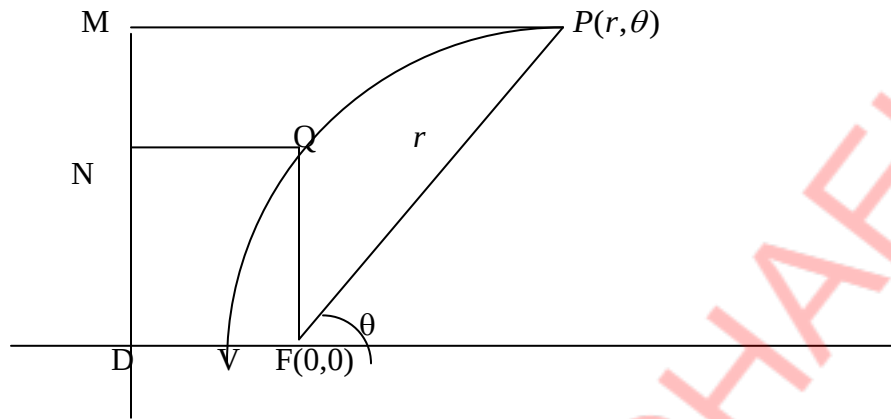


Figure 7.7

Let $P(r, \theta)$ be any point on the conic. Then, from the definition,

$$FP = ePM, \Rightarrow PM = r/e.$$

But also

$$PM = FD + r \cos \theta = r/e.$$

This gives

$$r = \frac{eFD}{1 - e \cos \theta}.$$

But also,

$$eFD = FQ = a,$$

where a is half the length of the latus rectum.

which implies

$$r = \frac{a}{1 - e \cos \theta}.$$

The conic can be distinguished from the polar equation by the value of e . If $e < 1$, the conic is an ellipse. If $e = 1$, the conic is a parabola. If $e > 1$, the conic is hyperbola.

Example 7.5

Consider the polar equation is

$$r = \frac{5}{4 - 3 \cos \theta}.$$

Writing it in standard form gives

$$r = \frac{5/4}{1 - (3/4) \cos \theta}.$$

Since e is less than one, the equation is that of an ellipse. The length of the latus rectum is $2a = 2 \times 5/4 = 5/2$.

Consider now the case when the directrix is on the right of the focus.

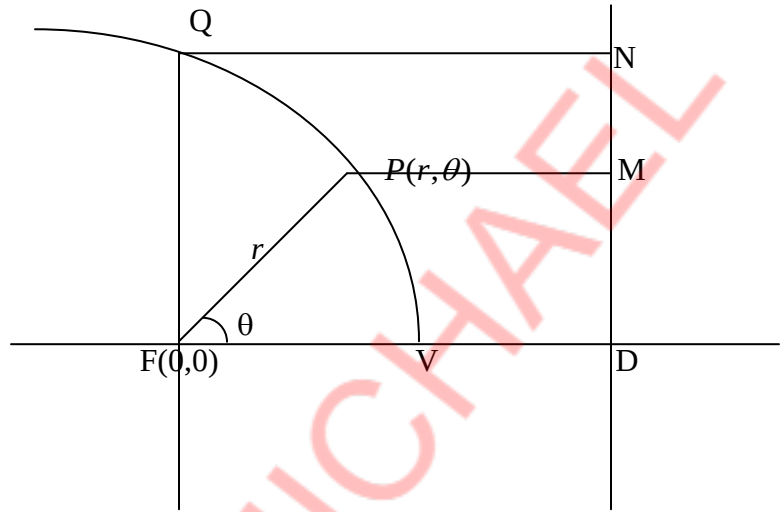


Figure 7.8

Again, let $P(r, \theta)$ be any point on the conic. Then, from the definition,

$$FP = ePM, \Rightarrow PM = r/e.$$

But also

$$PM = FD - r \cos \theta = r/e.$$

This gives

$$r = \frac{eFD}{1 + e \cos \theta}.$$

which implies

$$r = \frac{a}{1 + e \cos \theta}.$$

Using a similar argument, it can easily be shown that if the directrix is horizontal, then the equation is given by

$$r = \frac{a}{1 + e \sin \theta}$$

if the focus is at the origin and the directrix below the x -axis. If the focus is at the origin and the directrix above the x -axis, the polar equation of the conic is

$$r = \frac{a}{1 - e \sin \theta}$$

Example 7.6 Consider the conic equation $r = \frac{8}{3 + 9 \sin \theta}$. Writing the equation in standard form gives

$$r = \frac{8/3}{1 + 3\sin\theta}.$$

This is the equation of a hyperbola with focus at the origin, directrix parallel to and above the x -axis, and the length of the latus rectum being $2a = 2 \times 8/3 = 16/3$.

Exercises 7.7

Determine the kind of conic represented by the given polar equation, and give the eccentricity. Also state the position of the directrix in relation to the coordinate axis.

1. $r = \frac{1}{2 + \cos\theta}.$
2. $r = \frac{9}{33 + 11\cos\theta}.$
3. $r = \frac{7}{8 - \sin\theta}.$
4. $r = \frac{12}{2 + 15\sin\theta}.$
5. $r = \frac{14}{6 - 5\cos\theta}$

7.6 Converting a Cartesian equation to polar equation and vice versa.

We found that for point $P(x, y)$ whose polar coordinates are $P(r, \theta)$, then

$$x = r\cos\theta,$$

$$y = r\sin\theta,$$

$$\frac{y}{x} = \tan\theta,$$

$$r^2 = x^2 + y^2.$$

We use the relationships to transform any polar equation to Cartesian equation and vice versa.

Example 7.8

Write in Cartesian form the equation $r = 9\cos\theta$.

Solution

Put $r^2 = x^2 + y^2$ and $\cos\theta = \frac{x}{r} = \frac{x}{\sqrt{x^2 + y^2}}$

Hence

$$\sqrt{x^2 + y^2} = 9 \frac{x}{\sqrt{x^2 + y^2}} \Rightarrow x^2 + y^2 = 9x.$$

Or

$$(x - 4.5)^2 + y^2 = 4.5^2$$

which is the equation of the circle with center (4.5, 0) and radius 4.5.

Example 7.9

Write in Cartesian form of the equation $r = \frac{3}{2 - 2\cos\theta}$.

Solution

We get

$$\sqrt{(x^2 + y^2)} \left(2 - 2 \frac{x}{\sqrt{(x^2 + y^2)}} \right) = 3$$

$\Rightarrow$

$$2\sqrt{(x^2 + y^2)} = 2x + 3$$

$\Rightarrow$

$$4(x^2 + y^2) = 4x^2 + 12x + 9$$

This reduces to

$$y^2 = 3(x + 3/4).$$

This is the equation of a parabola with vertex $V(-3/4, 0)$, directrix $x = -3/2$, and length of latus rectum 6.

Example 7.10

Find the polar equation of the parabola $x^2 = -12(y - 3)$.

Solution

We have

$$r^2 \cos^2 \theta = -12(r \sin \theta - 3)$$

$$r^2 - r^2 \sin^2 \theta + 12r \sin \theta - 36 = 0$$

This reduces to

$$r = \frac{6}{1 + \sin \theta}$$

This is the polar equation of a parabola with focus at the origin, directrix $y = 6$.

It is now possible to get the polar equations of the general conic from the technique of conversion of a Cartesian equation to a polar equation. This is demonstrated in the examples below.

Example 7.11

Find the polar equation of the parabola $y^2 = 4ax$.

Solution

Put $r^2 = x^2 + y^2$ and $\cos \theta = \frac{x}{r} = \frac{x}{\sqrt{x^2 + y^2}}$.

This implies

$$r^2(1 - \cos^2 \theta) = 4ar \cos \theta$$

Example 7.12

Find the polar equation of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Solution

Put $r^2 = x^2 + y^2$ and $\cos \theta = \frac{x}{r} = \frac{x}{\sqrt{x^2 + y^2}}$
 $x^2 = r^2 \cos^2 \theta$ $y^2 = r^2 \sin^2 \theta$

Hence

$$\frac{r^2 \cos^2 \theta}{a^2} + \frac{r^2 \sin^2 \theta}{b^2} = 1$$

Or

$$r^2 [b^2 \cos^2 \theta + a^2 (1 - \cos^2 \theta)] = a^2 b^2$$

Or

$$r^2 (b^2 + (b^2 - a^2) \cos^2 \theta) = a^2 b^2.$$

This gives

$$r^2 = \frac{a^2 b^2}{a^2 - (a^2 - b^2) \cos^2 \theta} = \frac{a^2 b^2}{a^2 - a^2 e^2 \cos^2 \theta} = \frac{b^2}{1 - e^2 \cos^2 \theta}$$

e is eccentricity of the ellipse.

Example 7.13

Find the polar equation of the ellipse $x^2 + 9y^2 = 9$

Solution

In standard form, the equation is

$$\frac{x^2}{9} + \frac{y^2}{1} = 1$$

Hence $a = 3$ and $b = 1$. This gives

$$e^2 = \frac{a^2 - b^2}{a^2} = 8/9$$

$$r^2 = \frac{1}{1 - \frac{8}{9} \cos^2 \theta}$$

Miscellaneous Exercises

1. Convert the Cartesian coordinates to polar coordinates
(a) (7,8) (b) (-3,2) (c) (1,1) (d) (-9,30)
2. Plot the given point and give two other polar representations of the point.

(a) $\left(4, \frac{\pi}{2}\right)$

(b) $\left(5, \frac{\pi}{3}\right)$

(c) $\left(6, 7\frac{\pi}{6}\right)$

(d) $(2, 2\pi)$

3. Find the distance between the two given points.

(a) $\left(7, \frac{\pi}{3}\right), \left(6, \frac{\pi}{4}\right)$

(b) $\left(12, \frac{\pi}{6}\right), (1, \pi)$

(c) $\left(-6, -\frac{\pi}{2}\right), \left(-5, -\frac{\pi}{4}\right)$

(d) $\left(10, \frac{\pi}{9}\right), \left(-10, -\frac{\pi}{5}\right)$

BY ANASTASIA NDUTA

4. Convert the equation to polar form.
 - (a) $5x - 7y = 0$
 - (b) $x^2 + y^2 = 5$
 - (c) $6x + 9y = 0$
 - (d) $5x^2 + 3y^2 = 15$
 - (e) $y^2 - 8x - 16 = 0$
 - (f) $4x^2 + 4y^2 - 16y + 15 = 0$
5. Convert the polar equation to rectangular form.
 - (a) $r = 3\sin \theta$
 - (b) $r = 5\cos \theta$
 - (c) $r^2 = \sin 2\theta$
 - (d) $r^2 - 2r(\cos \theta - \sin \theta) - 7 = 0$
 - (e) $r = \frac{4}{1 - \cos \theta}$
6. Find the lengths of the sides of a triangle whose vertices are $\left(2, \frac{\pi}{8}\right), \left(-3, \frac{\pi}{4}\right), \left(-5, \frac{\pi}{2}\right)$
7. Find the polar equation of the straight line through $(3, -30^\circ)$ and parallel to the x-axis.
8. Find the equation of the circle with center $(10, 60^\circ)$ and radius $r = 3$.
9. Find the equation of the circle with center $(1, -30^\circ)$ and passes through $(3, -30^\circ)$ and $(9, 45^\circ)$

Summary

In this chapter we have introduced the concept of polar coordinates. We have shown how to:

- (a) A point on the plane can be presented in polar coordinates.
- (b) How to change a polar equation to Cartesian and Cartesian to polar coordinates.
- (c) Derive the polar equation of line, circle, and those of conic sections, namely the parabola, ellipse and hyperbola.

Further Reading

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- 9 P.K. Jain and Khalil Ahmad. **Analytical Geometry of Two Dimensions**. Wiley Eastern Limited, 1983.